UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

Unit 1 · Partial Differentiation · all sessions (Class 001 – Class 011) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)

S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

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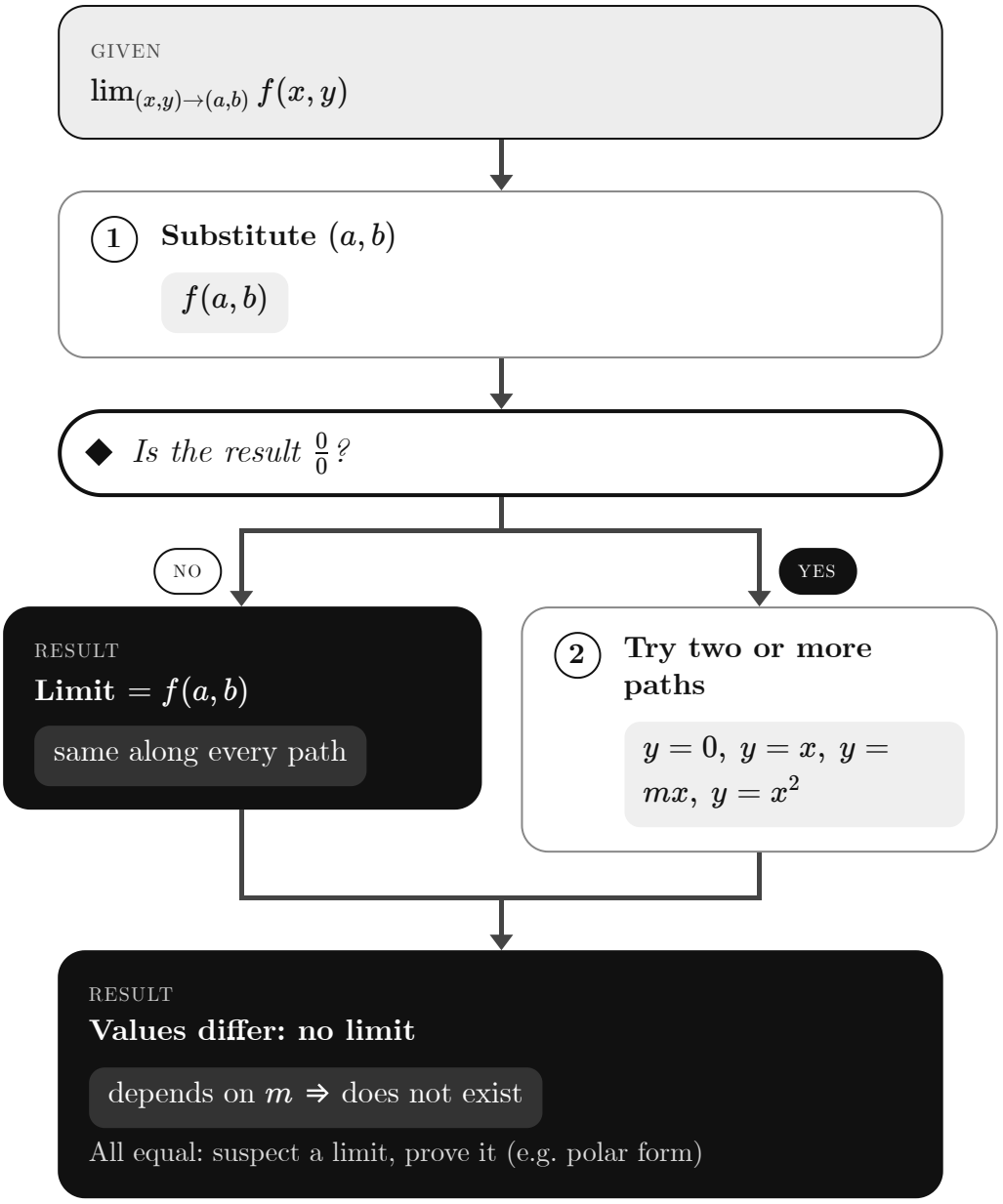
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38. One variable vs two variables
39. Total derivative vs chain rule vs implicit differentiation
40. Euler's theorem and its relatives
41. The extension for each outer function
42. Taylor's and Maclaurin's series side by side
43. Quick checks with a one-variable series
44. The four outcomes of the test
45. Stationary points of the two ESA functions
46. Elimination (class 010) and Lagrange's method

M1

Class 001 · Introduction to Partial Differentiation

2 Limit of a function of two variables



Remark. Agreement along a few paths is only evidence; disagreement is proof.

Exam trap. Read the point carefully: slide $(2,1)$ gives -8, the lecture's $(2,-1)$ gives -6.

Figure F2.1. Finding (or disproving) a limit at a point ESA ★★

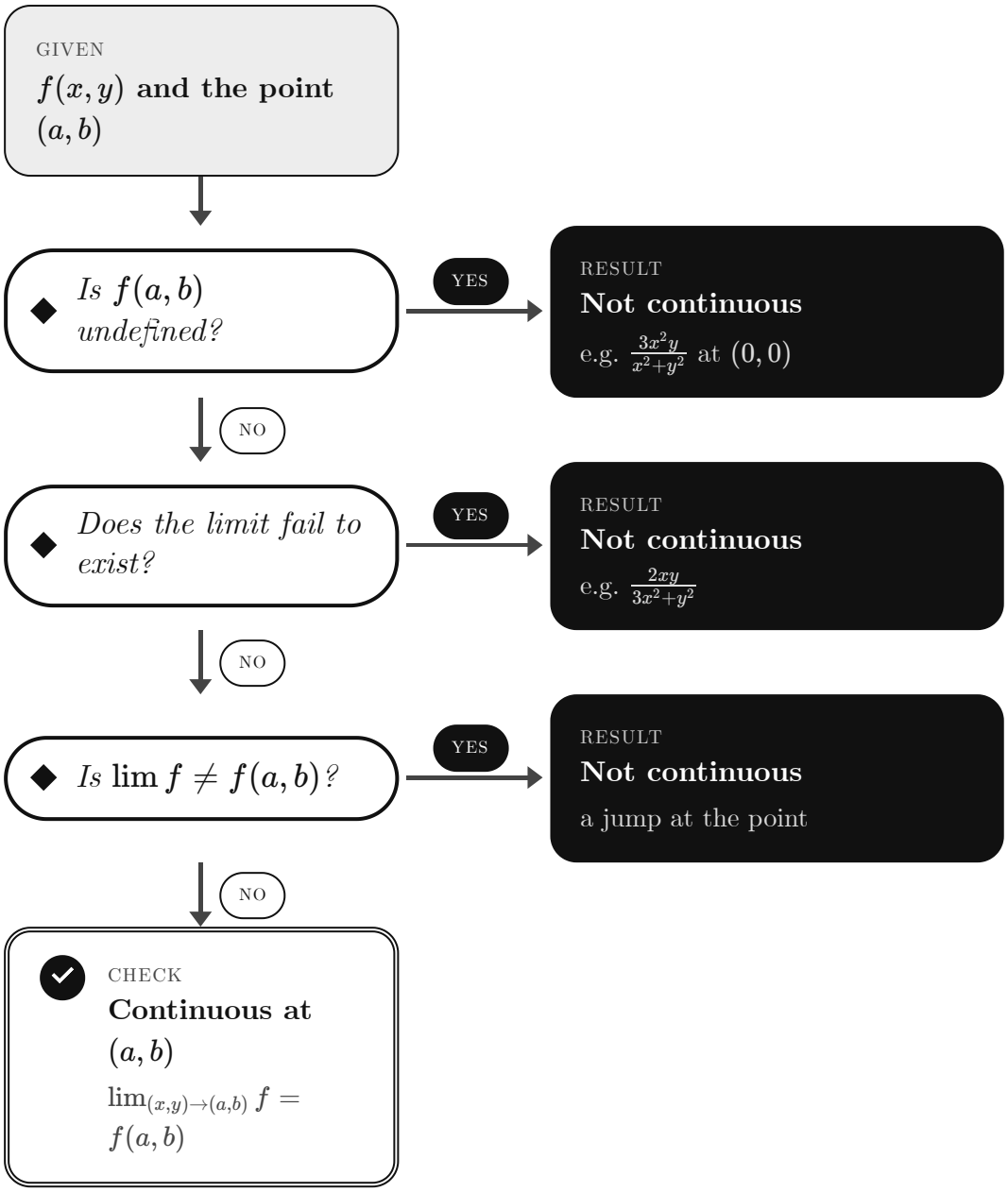
S001·9 · S001·10 · T001a 00:13:59 · T001a 00:17:42

One variable and two variables

	$y = f(x)$	$z = f(x, y)$
Domain	Real numbers	Ordered pairs (x, y)
Graph	A curve in the plane	A surface in space
Approach to the point	From the left or the right	Along infinitely many paths
Limit exists if	Left and right limits agree	Every path gives the same value
Continuous if	$\lim_{x \to c} f = f(c)$	$\lim_{(x,y) \to (a,b)} f = f(a, b)$
Derivative	$\frac{dy}{dx}$	$\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ (others constant)

S001·9 · S001·11 · T001a 00:09:54 · T001a 00:11:52

3 Continuity of a function of two variables



Remark. Only the first rung failing with the limit existing is a removable discontinuity: define $f(a,b)$ as the limit.

Figure F3.1. Is f continuous at (a, b)? ESA ★

S001·11 · T001a 00:18:46 · T001b 00:03:17

4 Partial derivatives: history, definition and notation

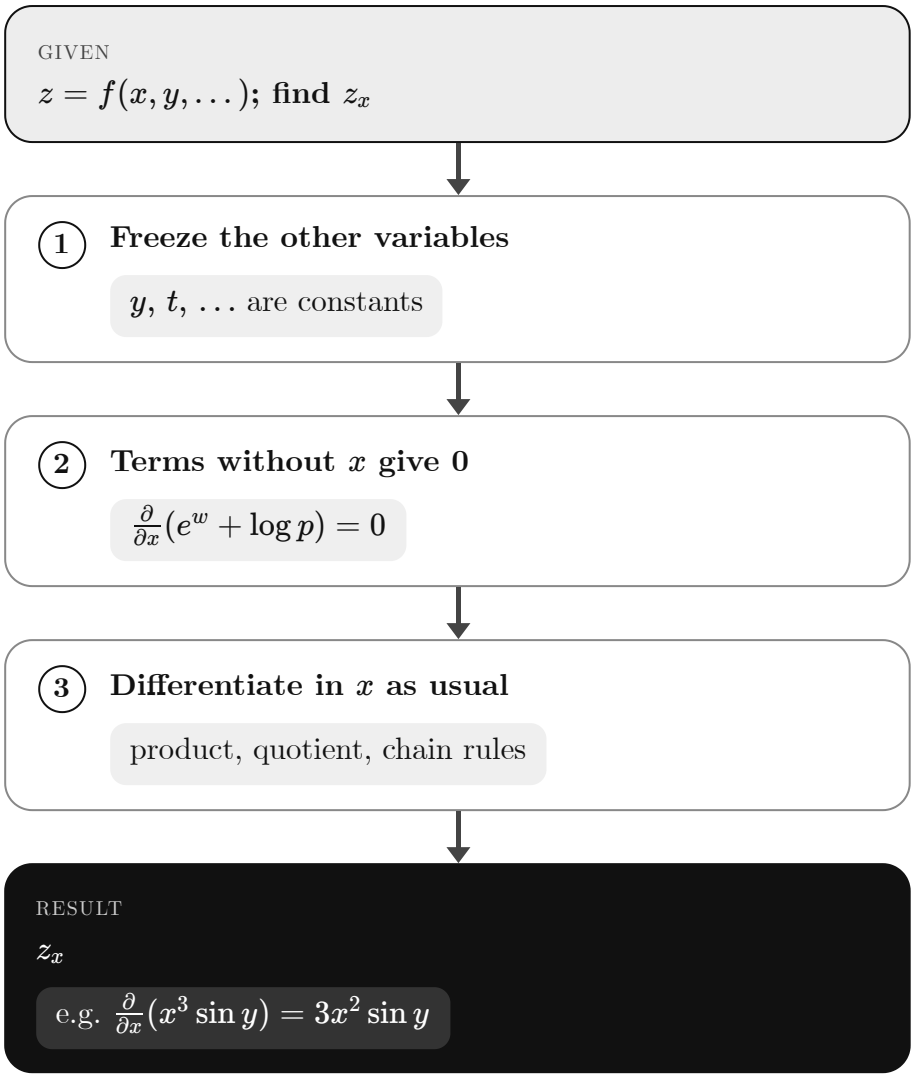
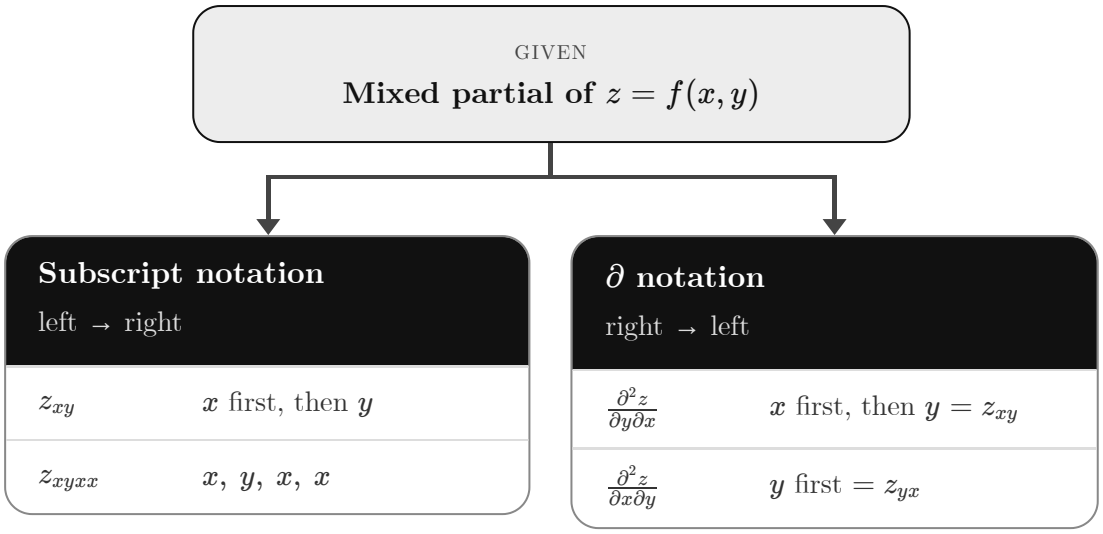


Figure F4.1. Computing a partial derivative ESA ★★

S001·15 · T001b 00:15:08 · T001b 00:13:57

5 Second-order and mixed partial derivatives



Remark. By Clairaut's theorem the order rarely changes the answer, but the notation is still examined.

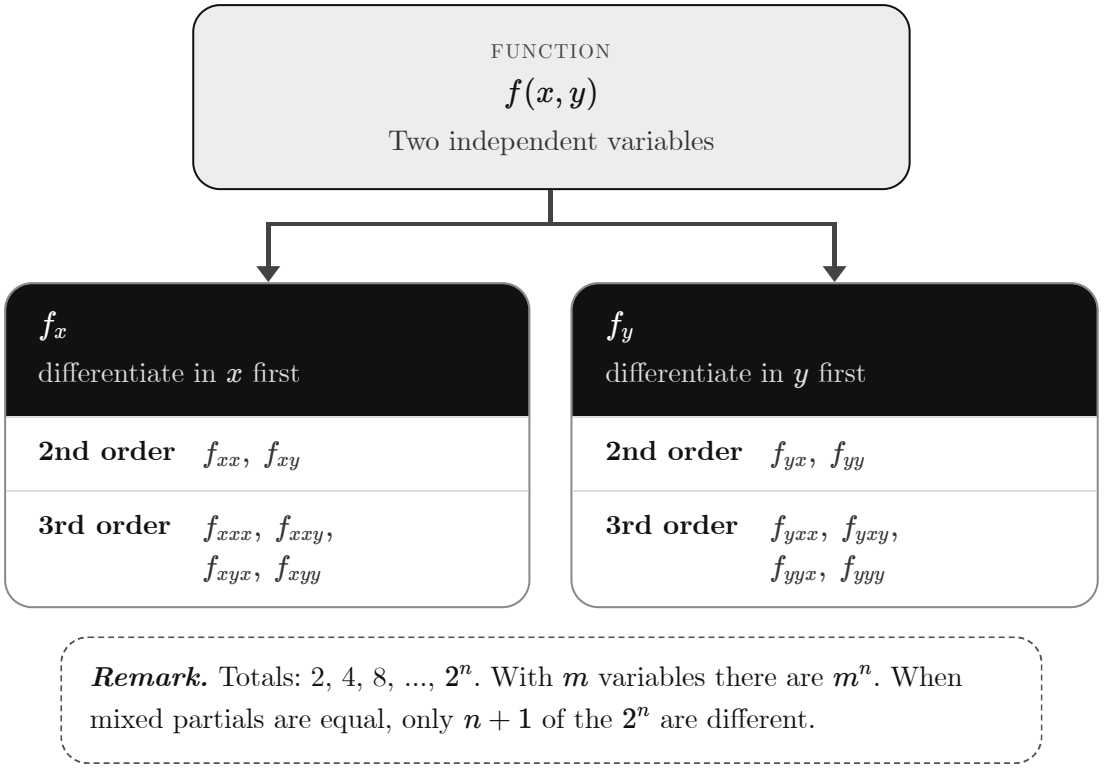
Figure F5.1. Reading the order of a mixed partial ESA ★★

S001·17 · T001b 00:18:51

M2

Class 002 · Geometrical Interpretation of Partial Derivatives

6 Higher-order partial derivatives

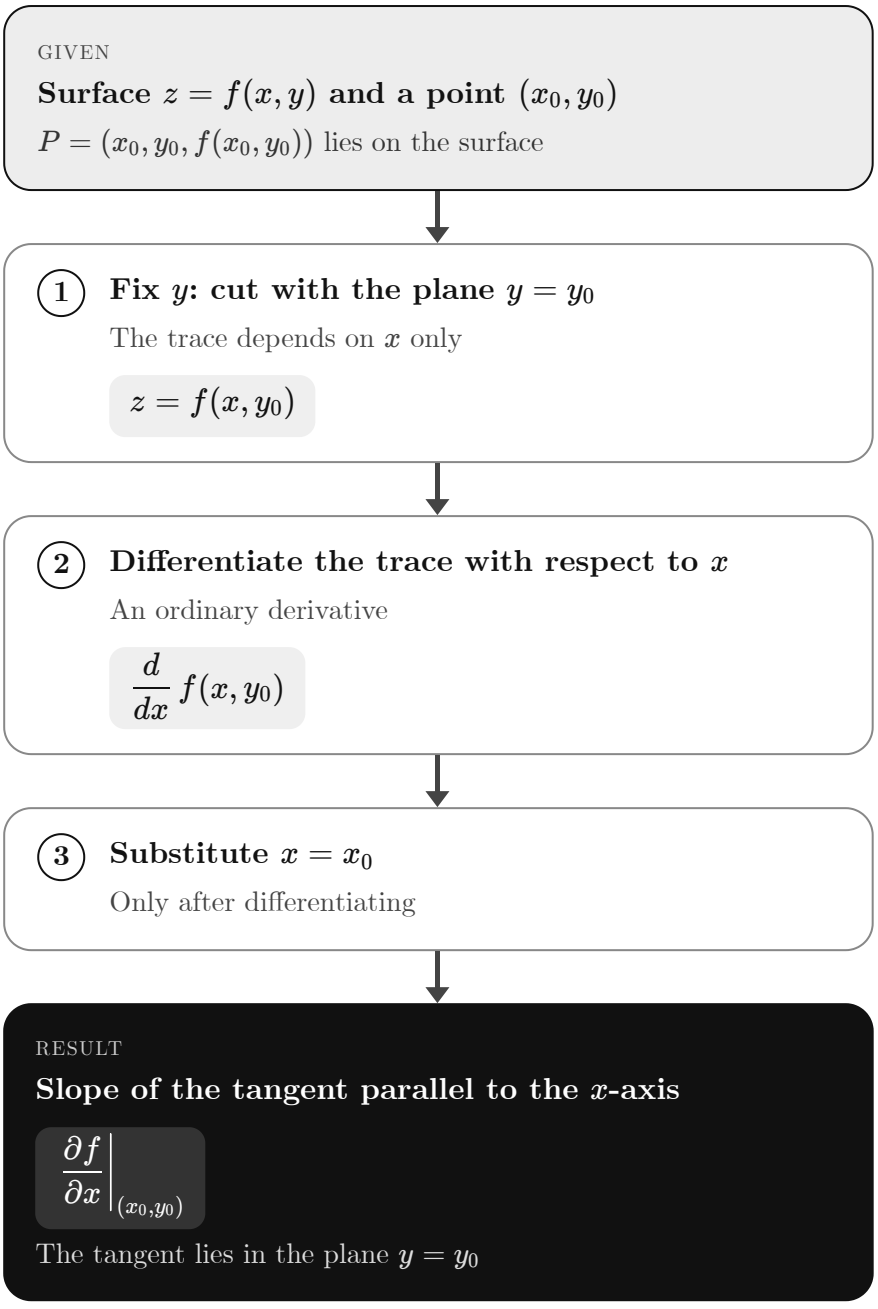


Remark. Totals: 2, 4, 8, ..., 2^n. With m variables there are m^n. When mixed partials are equal, only $n + 1$ of the 2^n are different.

Figure F6.1. Counting partial derivatives of each order. ESA ★

S002·3 · T002 00:01:37 · T002 00:02:13

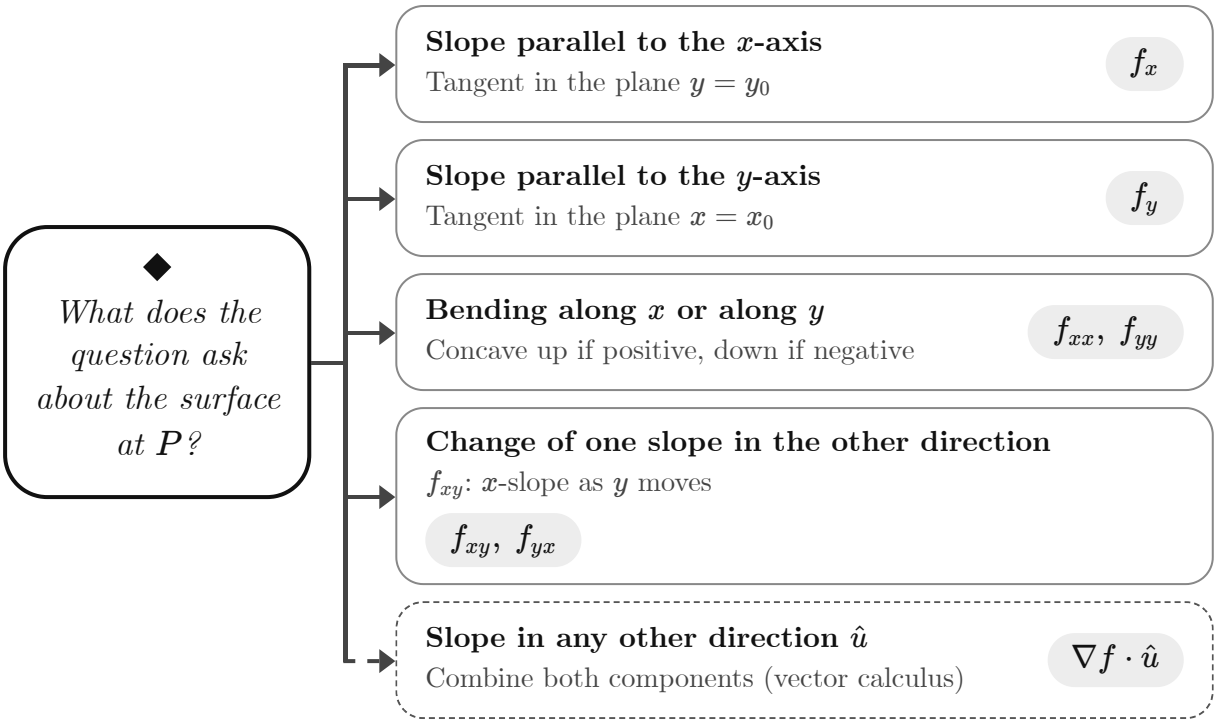
9 First-order partials as slopes of tangents



Remark. For $\partial f/\partial y$, swap the roles: cut with $x = x_0$, differentiate $f(x_0, y)$ in y, then put $y = y_0$.

Figure F9.1. Partial derivative as the slope of a tangent, step by step. ESA ★★

S002·6 · S002·7 · V002 00:07:45 · T002 00:06:10

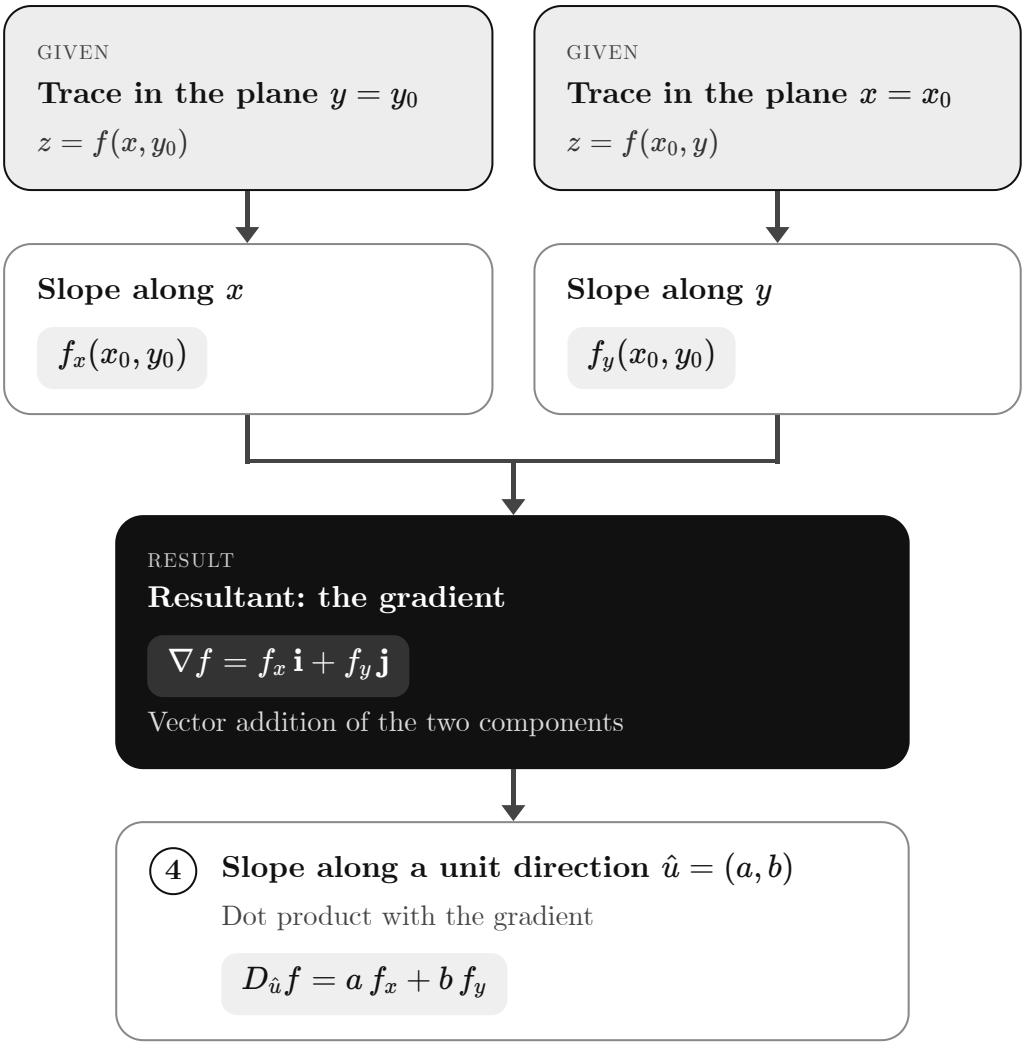


Remark. The dashed route is the gradient idea from slide 8. Its formula $\nabla f \cdot \hat{u}$ is added here for clarity and studied in vector calculus.

Figure F9.2. Which derivative answers which geometric question. ESA ★★

ADDED FOR CLARITY S002·6 · S002·8 · S002·9 · T002 00:07:54

10 Slope in any other direction: the gradient

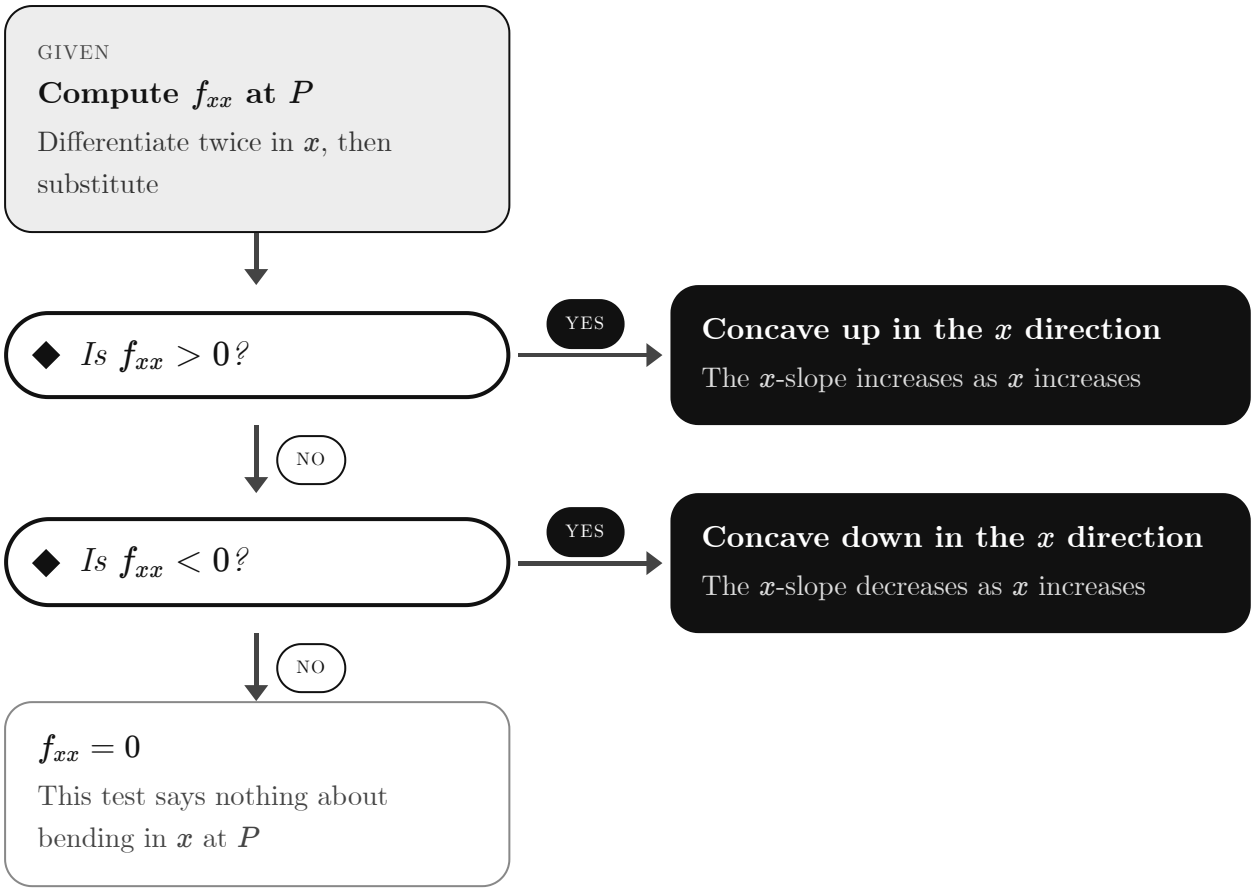


Remark. Check $\hat{u} = (1, 0)$: you get back f_x. The steepest slope is $|\nabla f| = \sqrt{f_x^2 + f_y^2}$. Example: the dome at $(1, -0.5)$ gives $\nabla f = (-2, 1)$ (W10.1).

Figure F10.1. Two slopes combine into the gradient. ESA ★

ADDED FOR CLARITY S002·8 · T002 00:09:06 · T002 00:09:23

11 Second-order partials: concavity and mixed partials



Remark. Run the same ladder with f_{yy} for the y direction. Read the sign of f_{xy} in the same way: positive means the x-slope increases as y increases (Figure G11.2).

Figure F11.1. Reading concavity from a second-order partial. ESA ★★

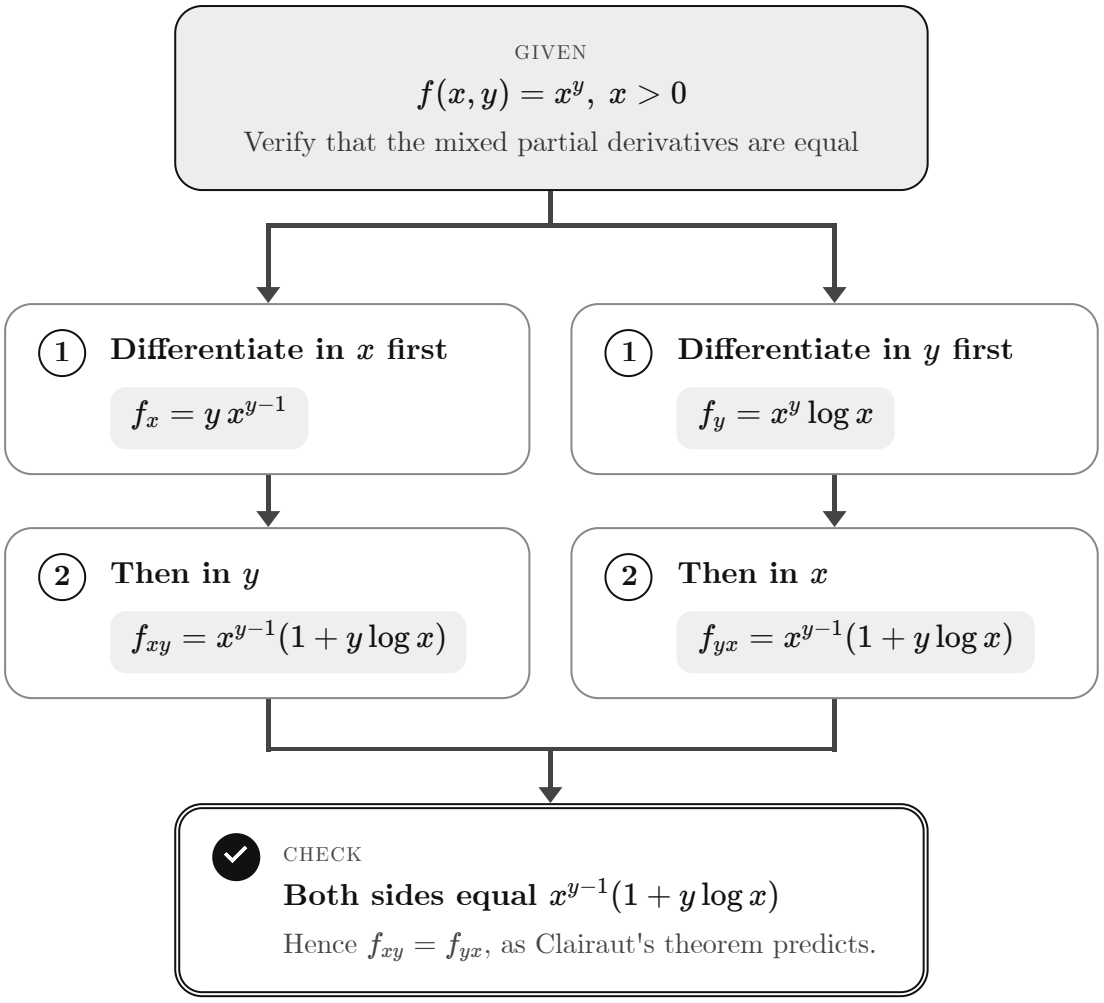
ADDED FOR CLARITY S002·9 · T002 00:11:19 · T002 00:11:38

One variable vs two variables

	One variable, $y = f(x)$	**Two variables, $z = f(x, y)$**
Graph	Curve in a plane	Surface in 3-D
Tangents at a point	One	Infinitely many, one per direction
First derivative	dy/dx: slope of the tangent	f_x, f_y: slopes parallel to the x- and y-axes; ∇f for other directions
Second derivative	d^2y/dx^2: concave up or down	f_{xx}, f_{yy}: concavity in x, y; f_{xy}, f_{yx}: change of one slope along the other direction
Number of order n	1	2^n (or m^n for m variables)

ADDED FOR CLARITY S002·4 · S002·6 · S002·9 · S002·3

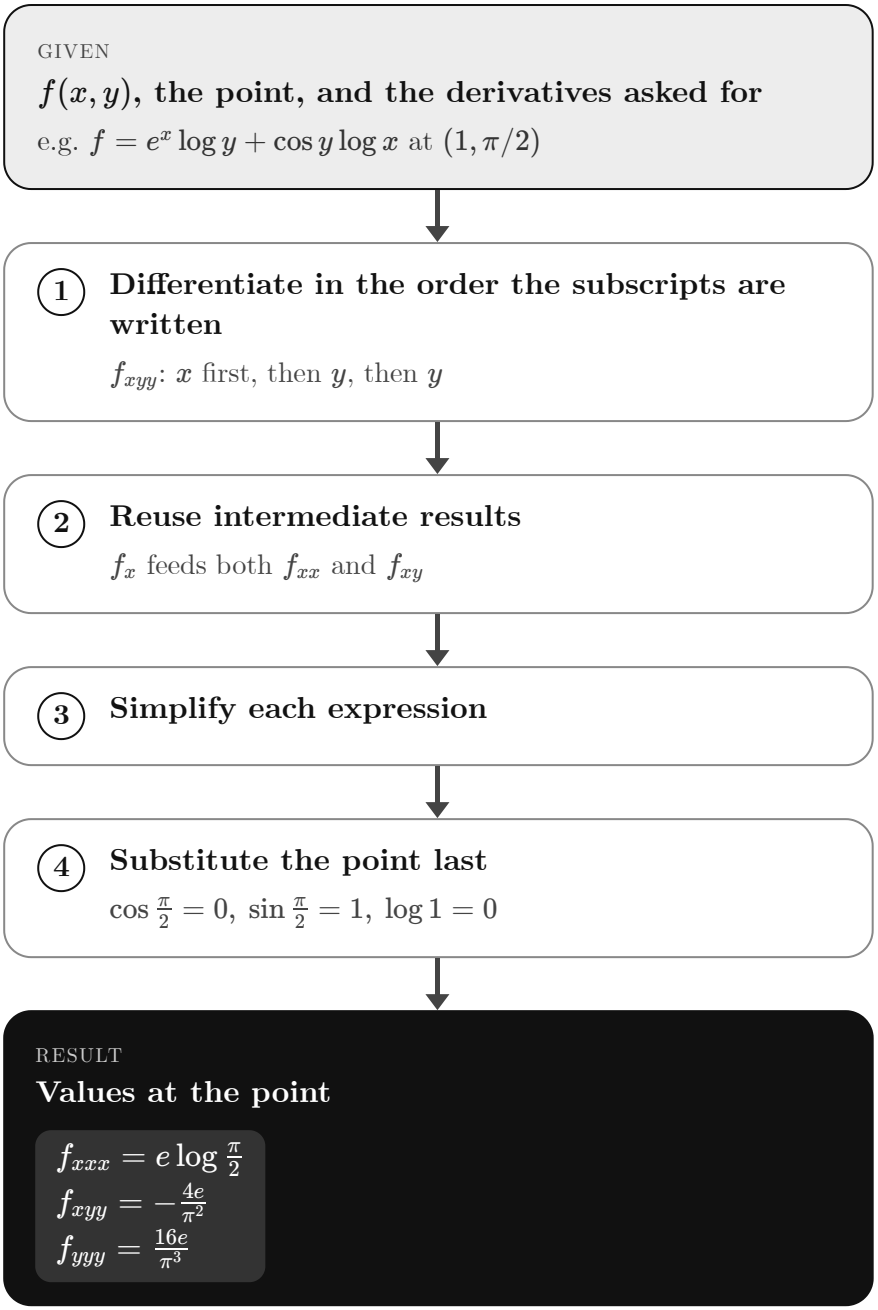
12 Problems on partial derivatives



Exam trap. In x, x^y is a power; in y, it is an exponential. Using the wrong rule is the usual slip.

Figure F12.1. Verifying equal mixed partial derivatives. ESA ★★

CW U1 P2 · T002 00:14:59



Remark. log is the natural logarithm, so $\frac{d}{dy}\log y = \frac{1}{y}$. Check one value against the CW answer key if you can.

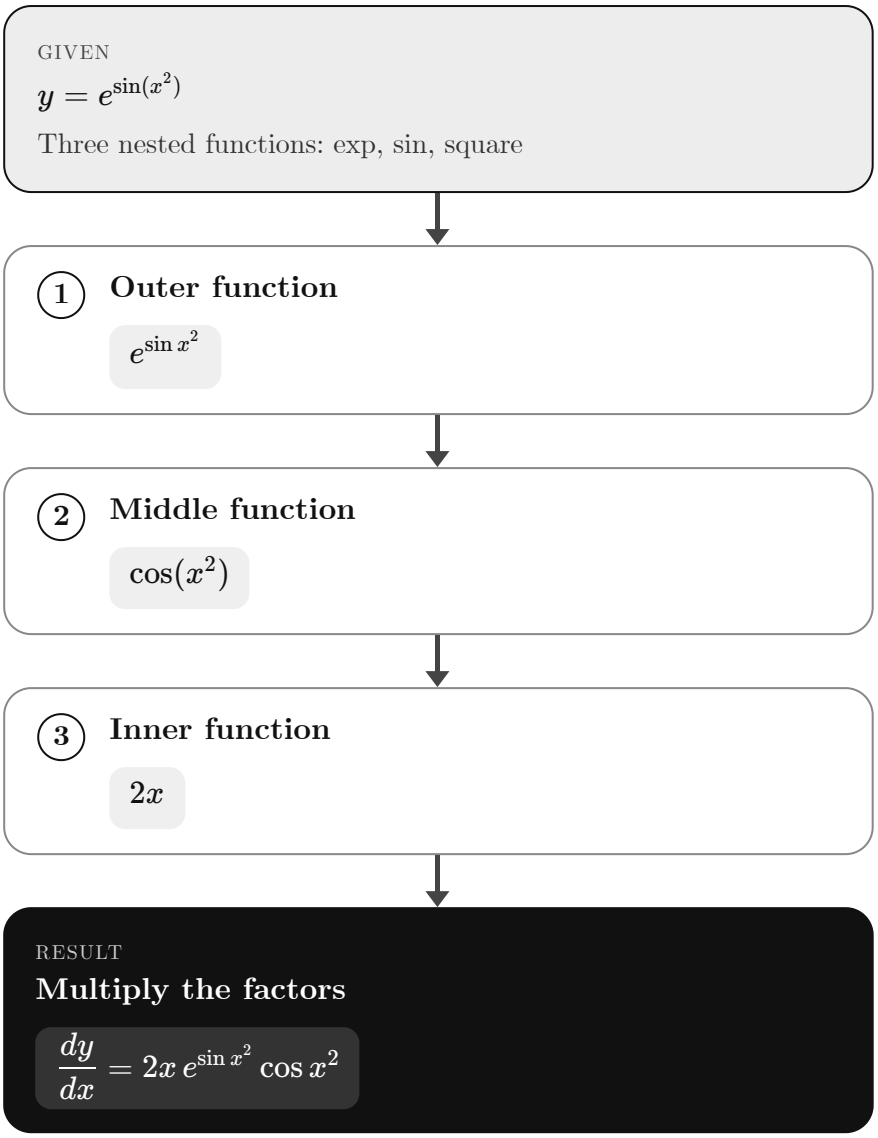
Figure F12.2. Evaluating higher-order partials at a point (ESA Dec 2023 Q1.a). ESA ★★

ESA Dec-23 1a · CW U1 P5 · S002·3

M3

Classes 003–004 · Total Derivative, Chain Rule and Implicit Functions

13 Composite functions and the one-variable chain rule

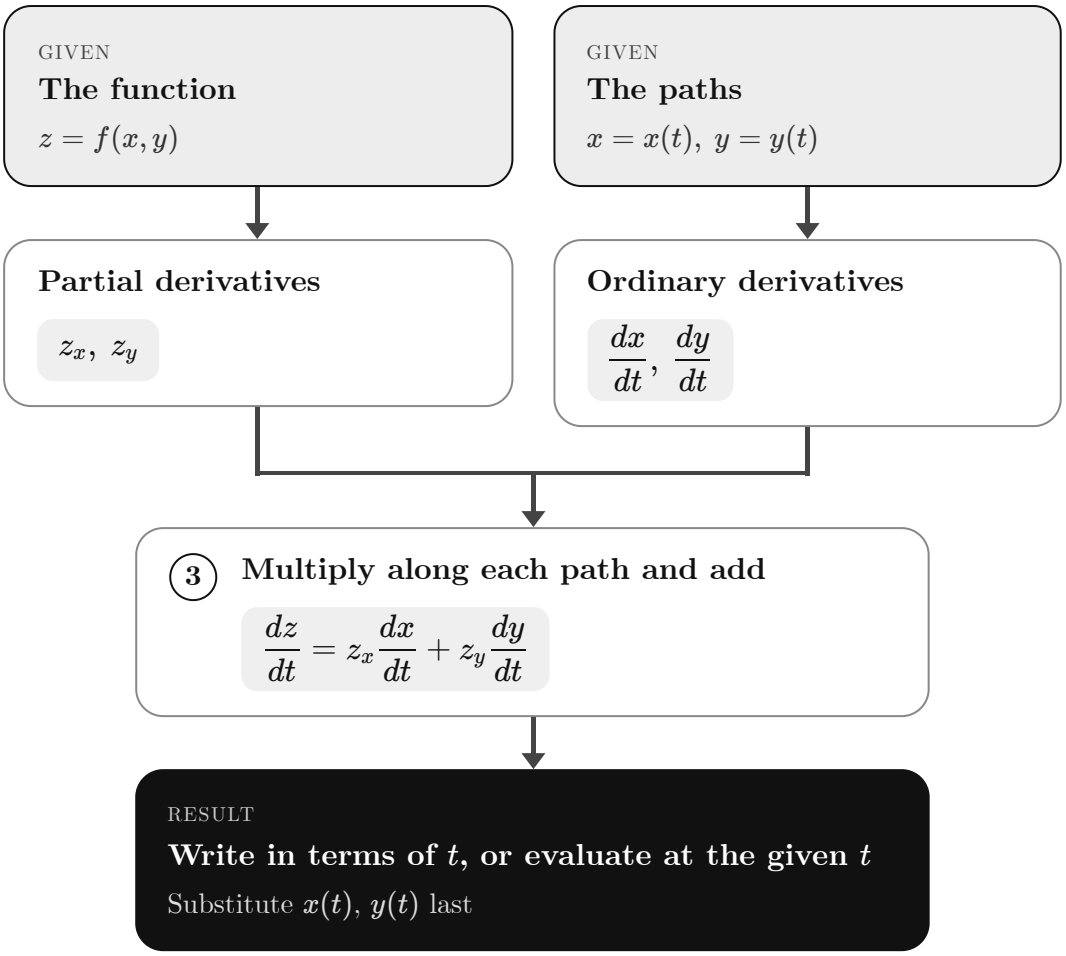


Remark. The multivariable rules in this pack work the same way, except that each intermediate variable adds its own term.

Figure F13.1. Differentiating a composite function from the outside in. ESA ★

S003·4 · T003a 00:00:33

14 The total derivative rule

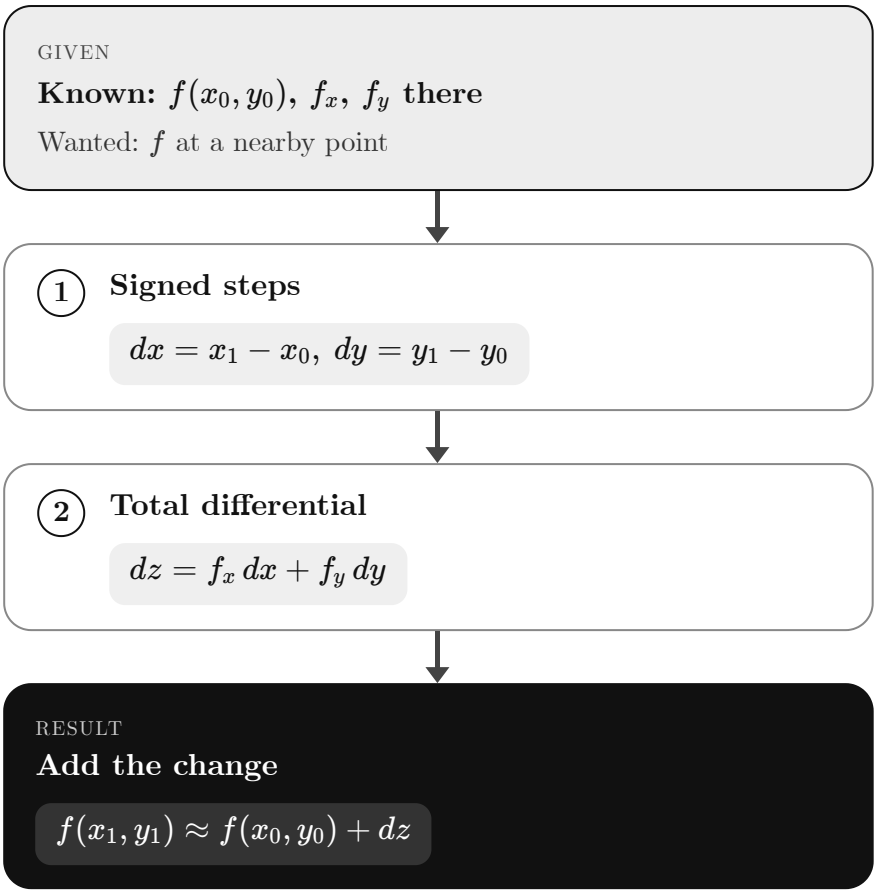


Remark. Check on a small example by substituting first: both routes must agree. On slide 003·6 they do not, which exposes the slide's error (W14.1).

Figure F14.1. Total derivative in three moves. ESA ★★

S003·5 · S003·18 · T003b 00:01:10

15 The total differential: approximation and sensitivity

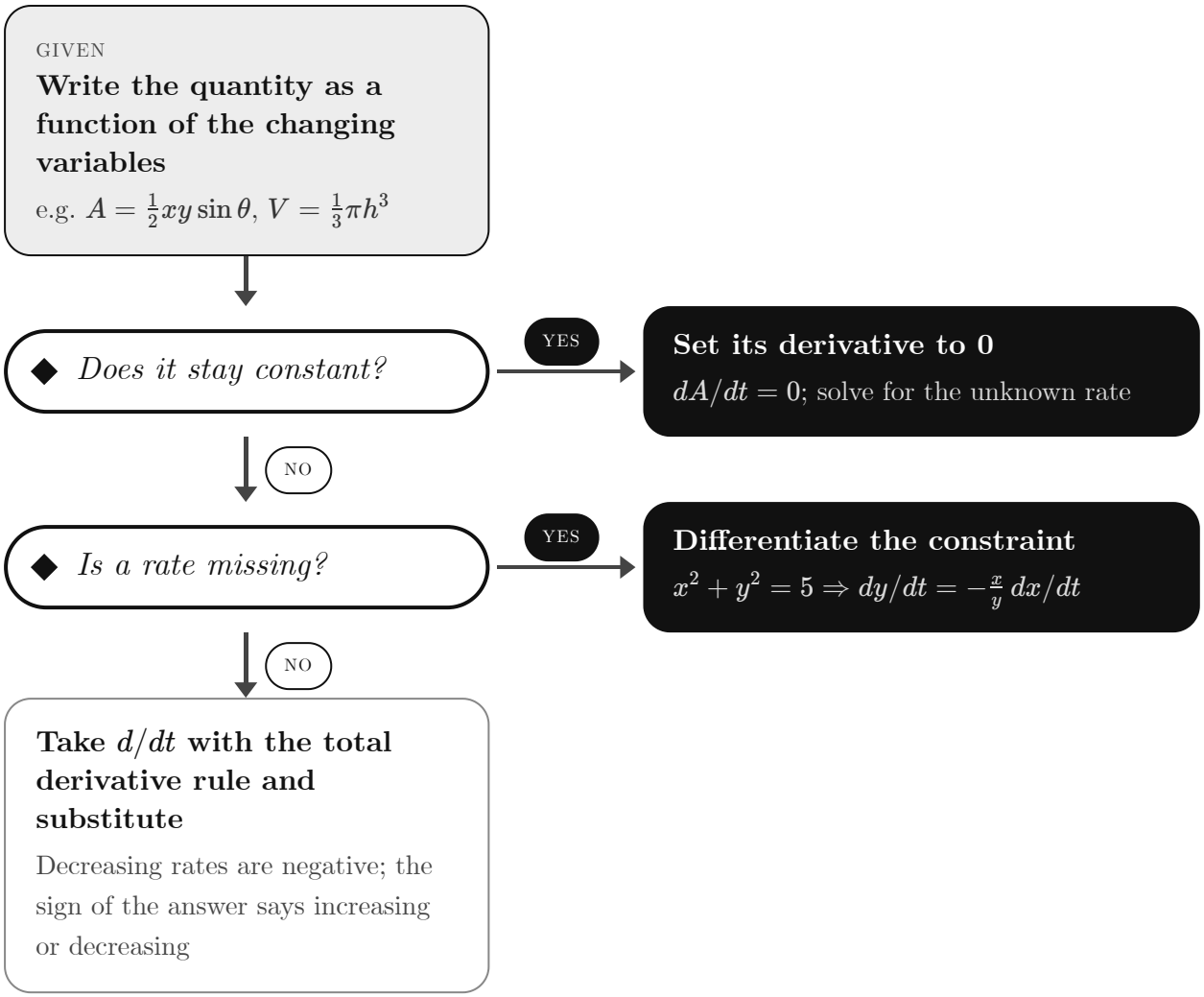


Remark. Slide 003·8: $dz = 1.3(0.1) + (-0.6)(-0.03) = 0.148$, so $f(2.1, -3.03) \approx 6.148$.

Figure F15.1. Estimating a value near a known point. ESA ★★

S003·8 · T003a 00:13:09

16 Rates of change problems

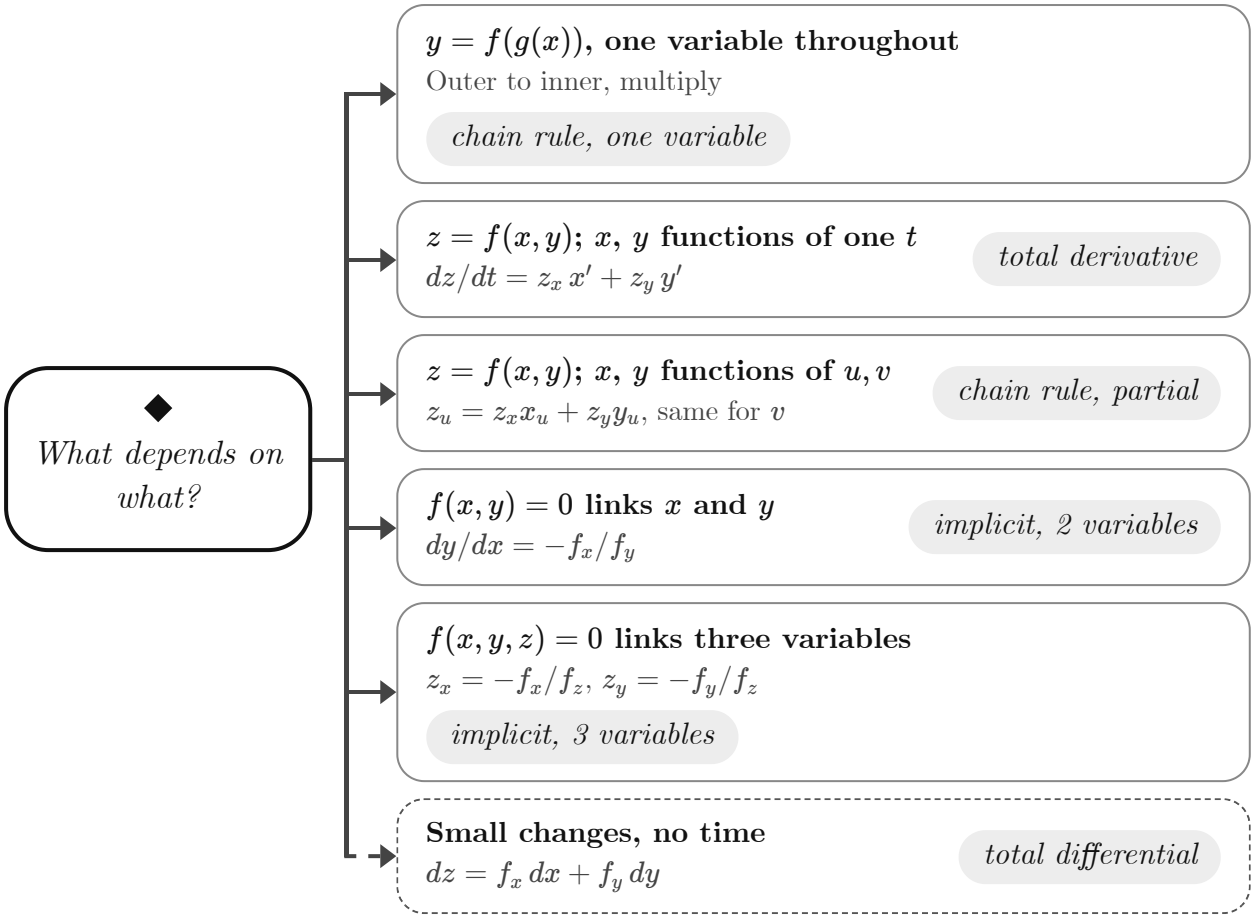


Remark. Both checks can apply in one problem; work down the ladder, then substitute the numbers last.

Figure F16.1. Solving a rates-of-change problem. ESA ★★

S003·7 · S003·9 · S003·21 · T003a 00:07:25

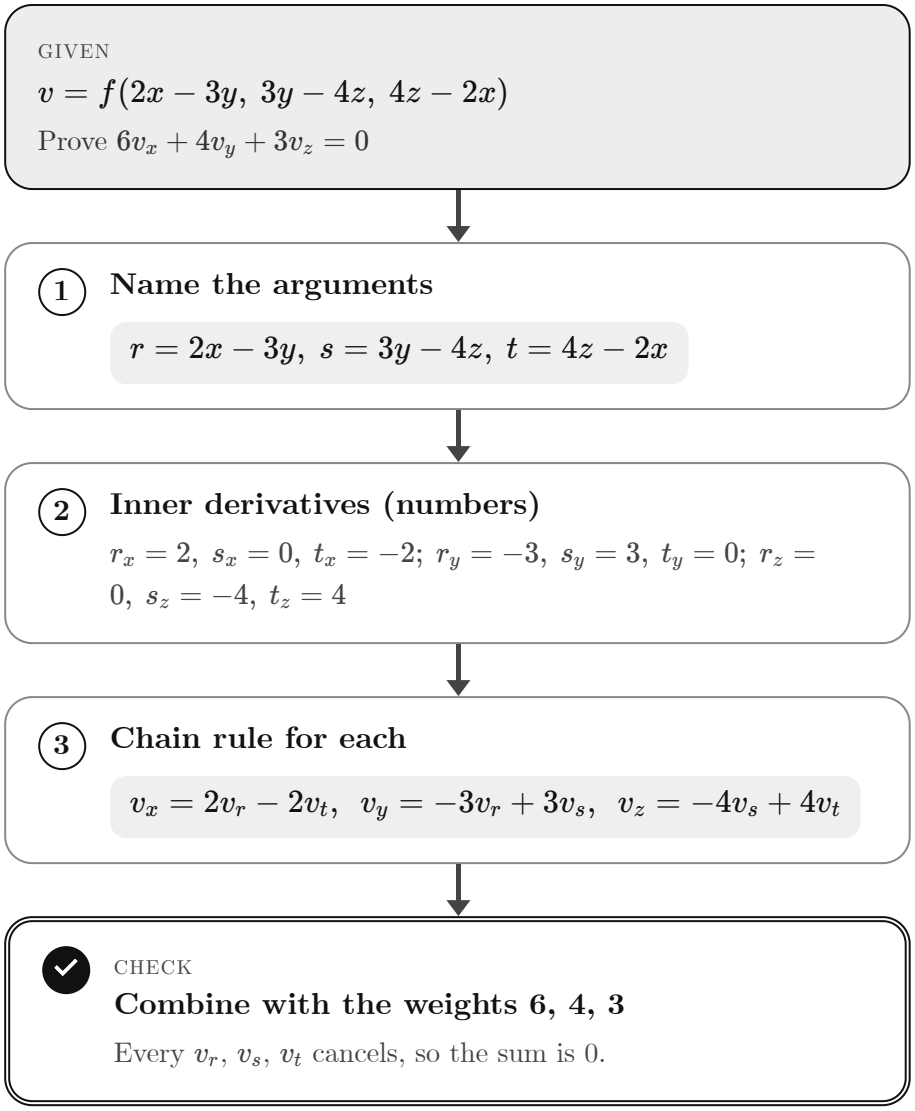
17 The chain rule for composite functions



Remark. The dashed route has no derivative with respect to a variable: it estimates a change. Everything else starts from the tree diagram.

Figure F17.1. Which rule? Read the chain first. ESA ★★★

S003·5 · S003·13 · S004·5 · S004·6 · T004b 00:02:31



Exam trap. Slide 003·16 drops the 3 in $v_y = -3v_r + 3v_s$; without it the terms do not cancel.

Figure F17.2. Proving an identity such as $6v_x + 4v_y + 3v_z = 0$. ESA ★★★

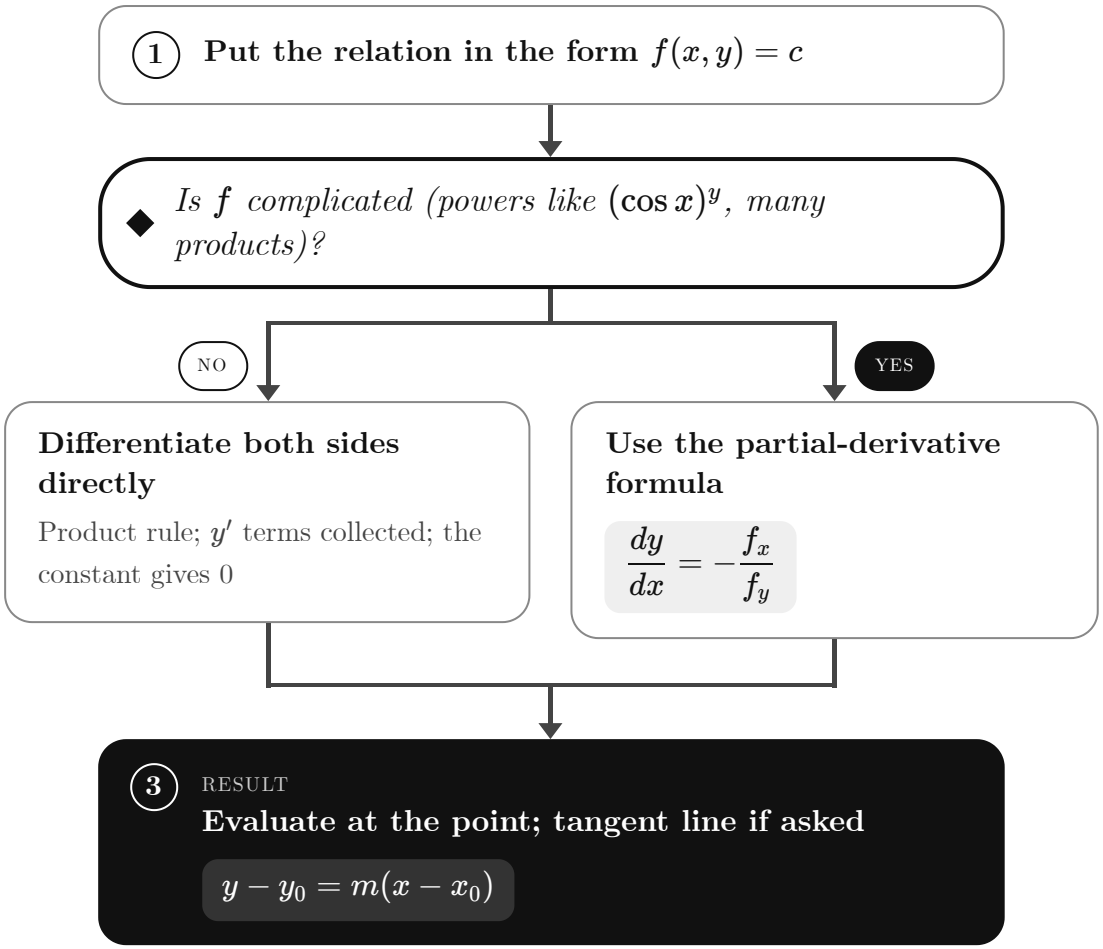
S003·15 · S003·16 · ESA Jan-24 s2 1d

Total derivative vs chain rule vs implicit differentiation

	Total derivative	Chain rule (partial)	Implicit function
Set-up	$z = f(x, y)$, $x(t)$, $y(t)$	$z = f(x, y)$, $x(u, v)$, $y(u, v)$	$f(x, y) = 0$ or $f(x, y, z) = 0$
Final variables	one: t	two: u, v	x (and y)
Formula	$z_x\, x' + z_y\, y'$	$z_x x_u + z_y y_u$	$-f_x/f_y$, $-f_x/f_z$
Result written	dz/dt	$\partial z/\partial u$, $\partial z/\partial v$	dy/dx, $\partial z/\partial x$
Typical question	rate of change at an instant	prove an identity	slope or tangent line

ADDED FOR CLARITY S003·5 · S003·13 · S004·4 · S004·6

18 Implicit functions of two variables

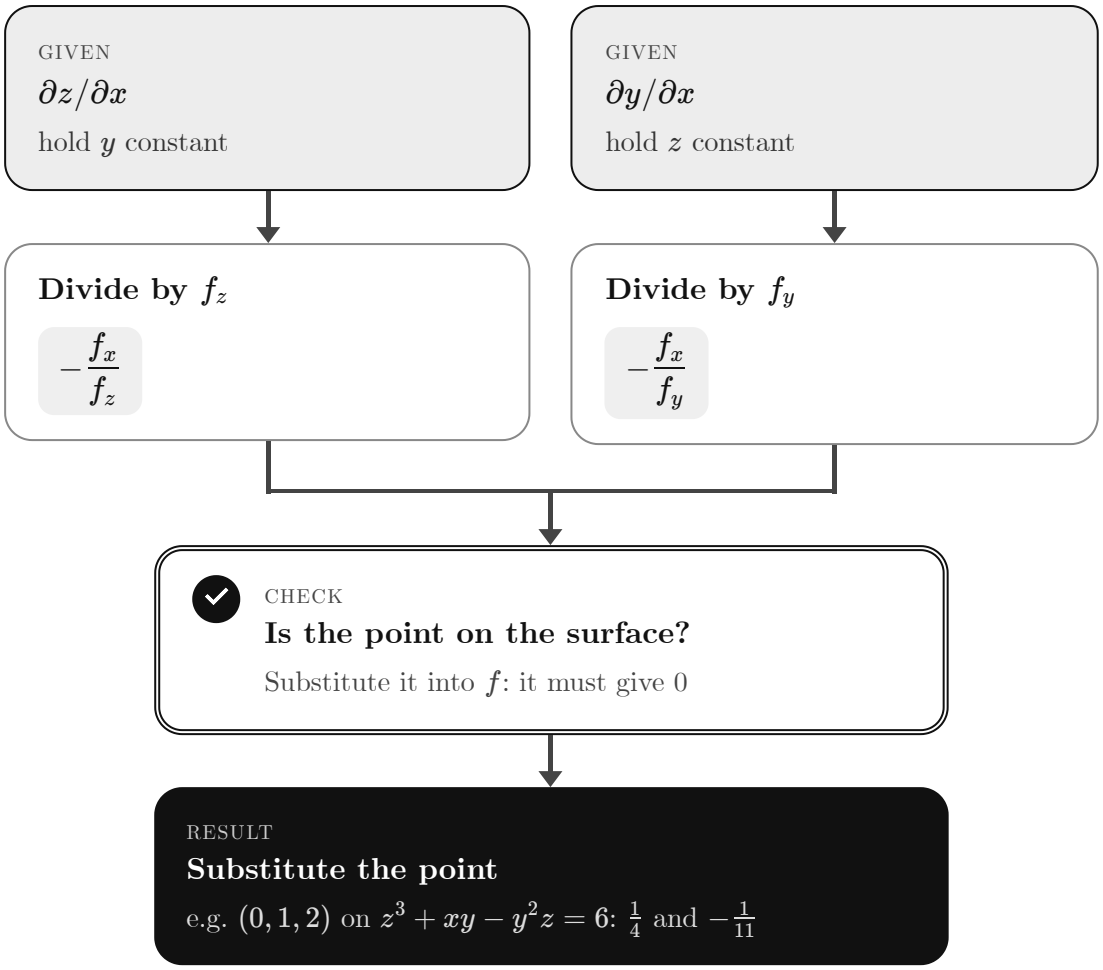


Remark. Both routes give the same dy/dx (W18.1, W18.2). The formula needs $f_y \neq 0$ at the point.

Figure F18.1. Finding dy/dx for an implicit function: two routes. ESA ★★

S003·27 · S004·3 · S004·5 · T003b 00:10:33 · T004a 00:06:19

19 Implicit functions of three variables



Exam trap. The lecture used $(1,0,2)$, which is not on the surface; always check the point first.

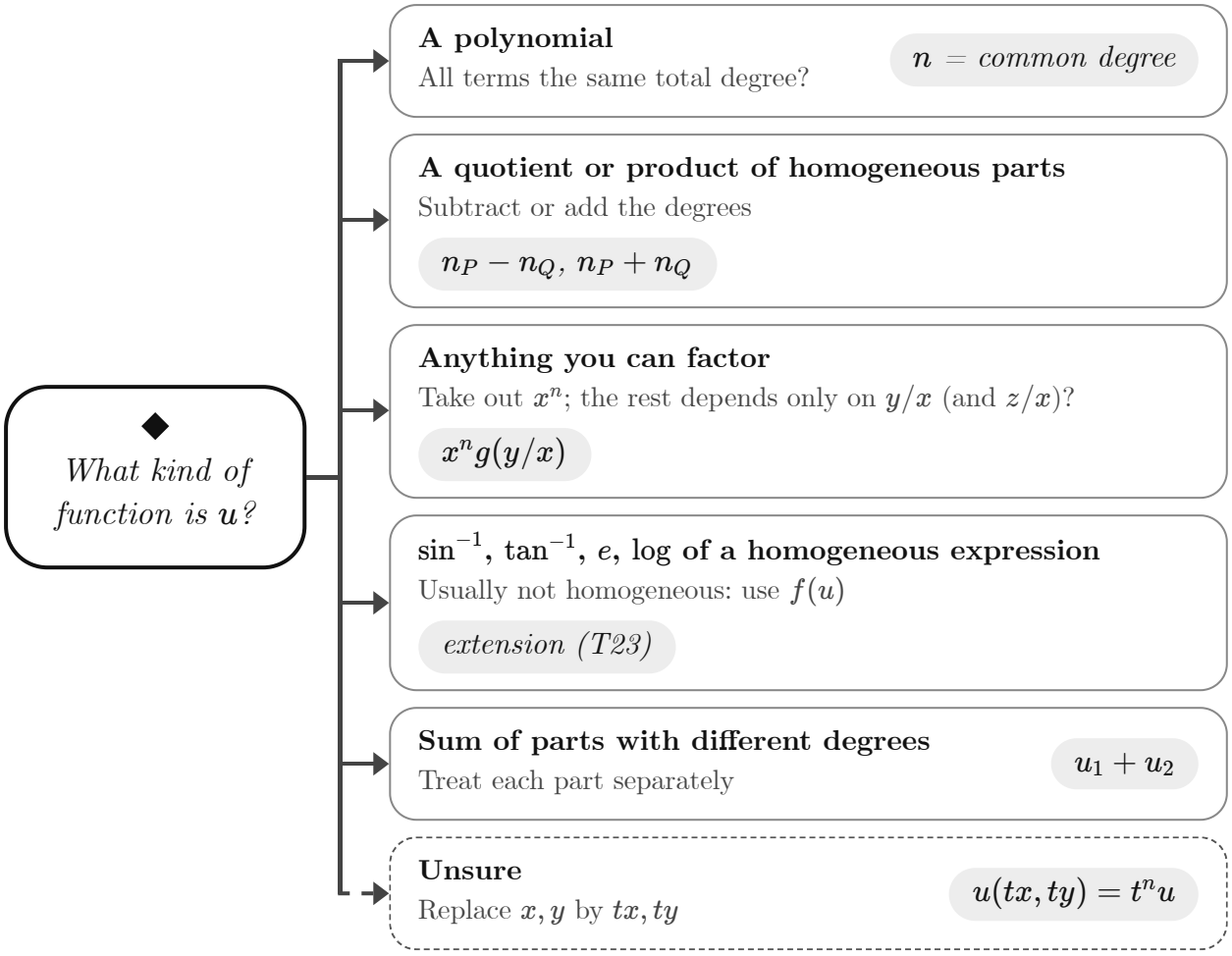
Figure F19.1. Partial derivatives from $f(x,y,z)=0$. ESA ★★

S004·6 · S004·9 · S003·26

M4

Class 005 · Homogeneous Functions and Euler's Theorem

20 Homogeneous functions



Remark. The dashed route (the scaling test) is added for clarity; it always works and confirms the degree found by factoring.

Figure F20.1. Is it homogeneous, and of what degree? ESA ★★

S005·3 · S005·4 · T005 00:02:36 · T005 00:05:42

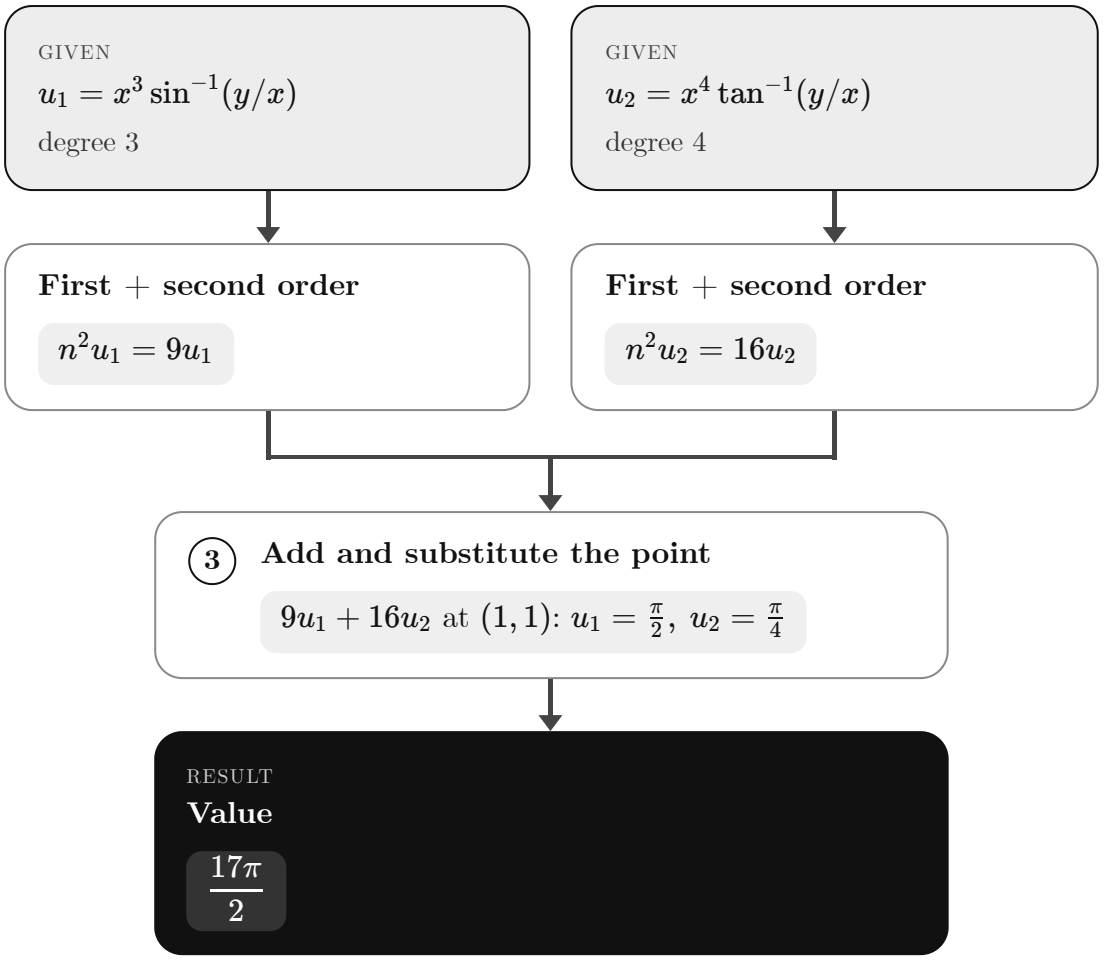
21 Euler's theorem

Euler's theorem and its relatives

	Applies when	Left side	Right side
Euler, first order	u homogeneous, degree n	$xu_x + yu_y$	nu
Corollary, second order	u homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$n(n-1)u$
Both added	u homogeneous, degree n	first + second order	n^2u
Extension, first order	$f(u)$ homogeneous, degree n	$xu_x + yu_y$	$g(u) = nf(u)/f'(u)$
Extension, second order	$f(u)$ homogeneous, degree n	$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$	$g(u)[g'(u) - 1]$
Three variables	$u(x, y, z)$ homogeneous	$xu_x + yu_y + zu_z$	nu

added for clarity S005·6 · S005·7

22 Second and higher order: the corollary



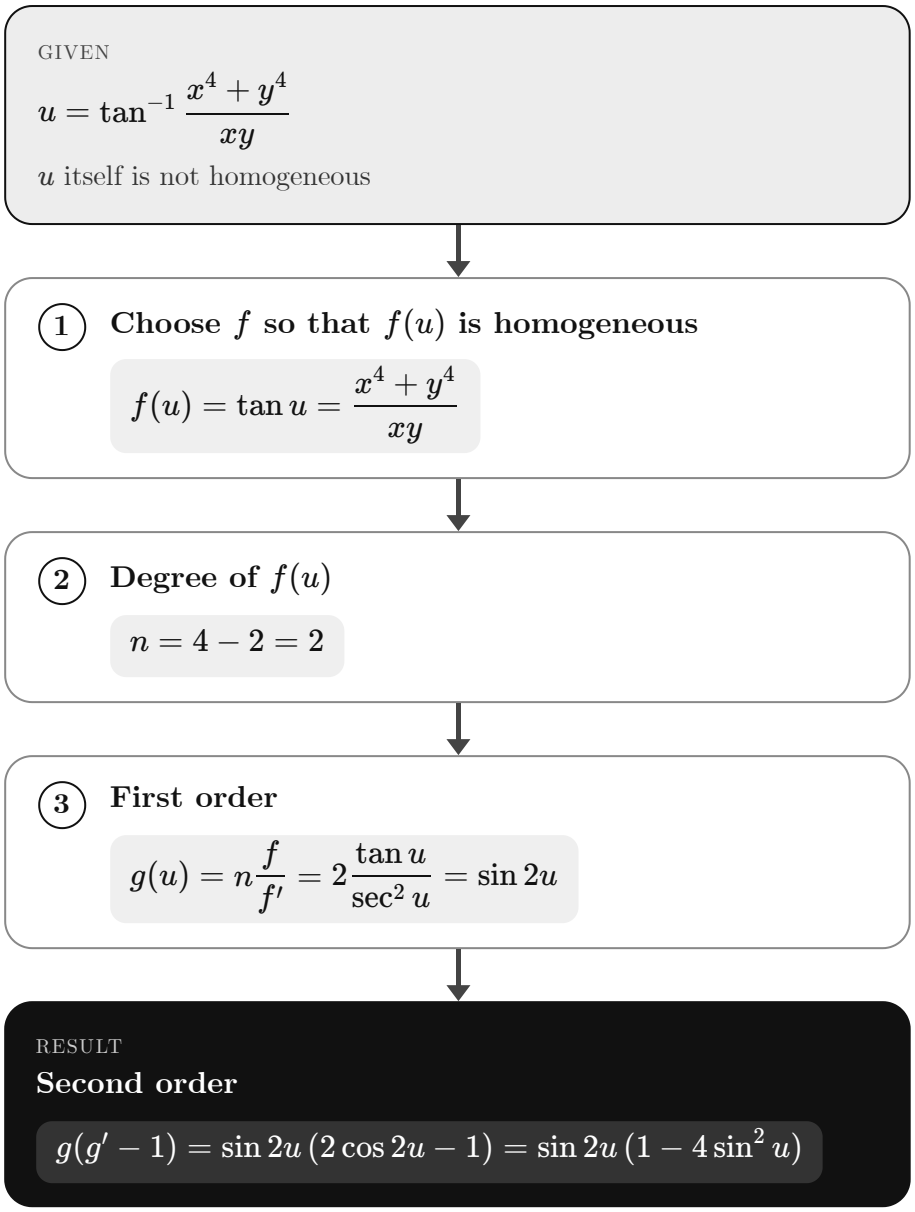
Remark. Only first order: n; only second order: $n(n-1)$; both: n^2. Read which combination the question asks for.

Figure F22.1. Evaluating Euler combinations for a sum of homogeneous parts.

ESA ★★★

ESA Jan-24 s3 1c · ESA Jan-24 s4 1d

23 Extension: when f(u) is homogeneous



Remark. Common choices: $\sin^{-1}(\dots) \to f = \sin$ ($g = n\tan u$); $\tan^{-1} \to f = \tan$ ($g = \frac{n}{2}\sin 2u$); $\log(\dots) \to f = e^u$ ($g = n$); $e^{(\cdots)} \to f = \log$ ($g = nu\log u$).

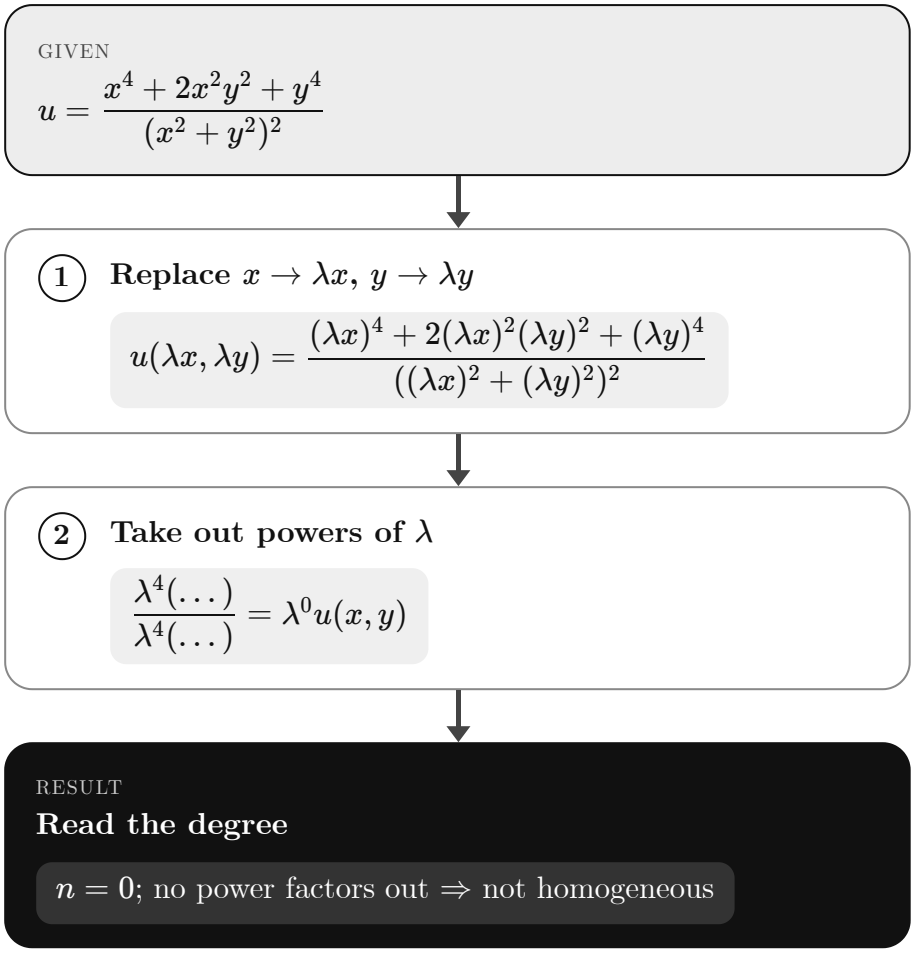
Figure F23.1. Using the extension of Euler's theorem. ESA ★★★

S005·7 · ESA May-24 1a

M5

Class 006 · Problems on Euler's Theorem

25 Testing homogeneity: the scaling test

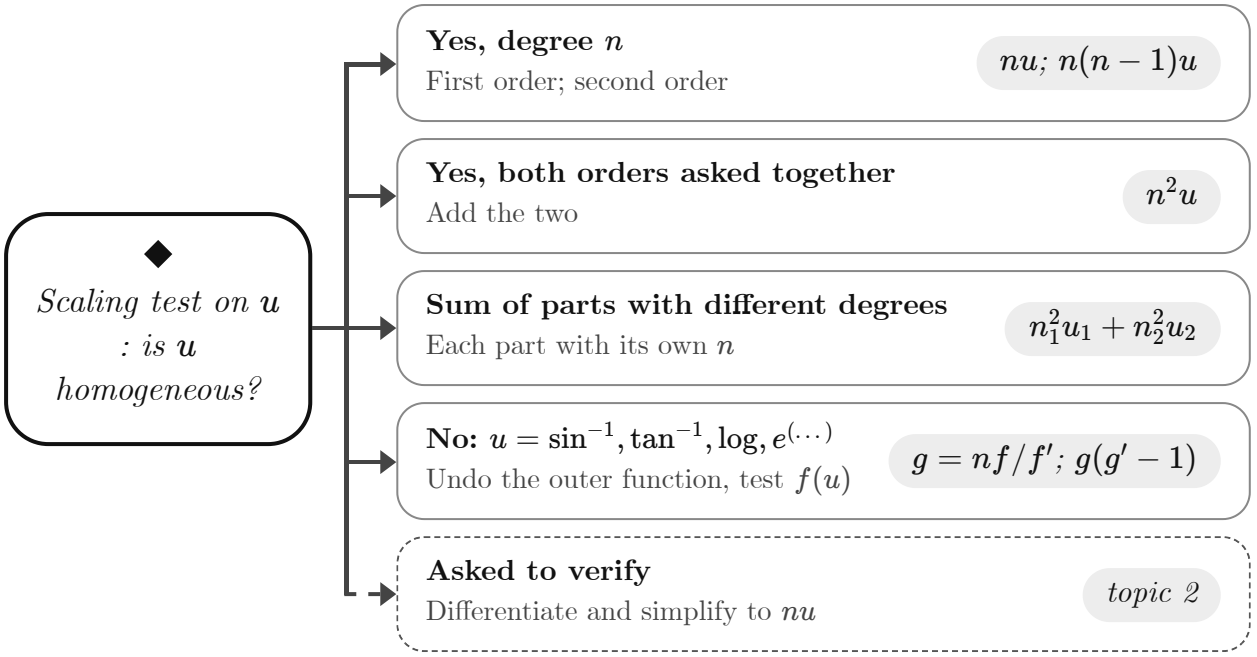


Remark. Use the test on u (topic 3) or on $f(u)$ (topic 4); the steps are the same.

Figure F25.1. The scaling test ESA ★★

S006·5 · T006b 00:01:41 · T006b 00:02:37

27 When u itself is homogeneous: first and second order

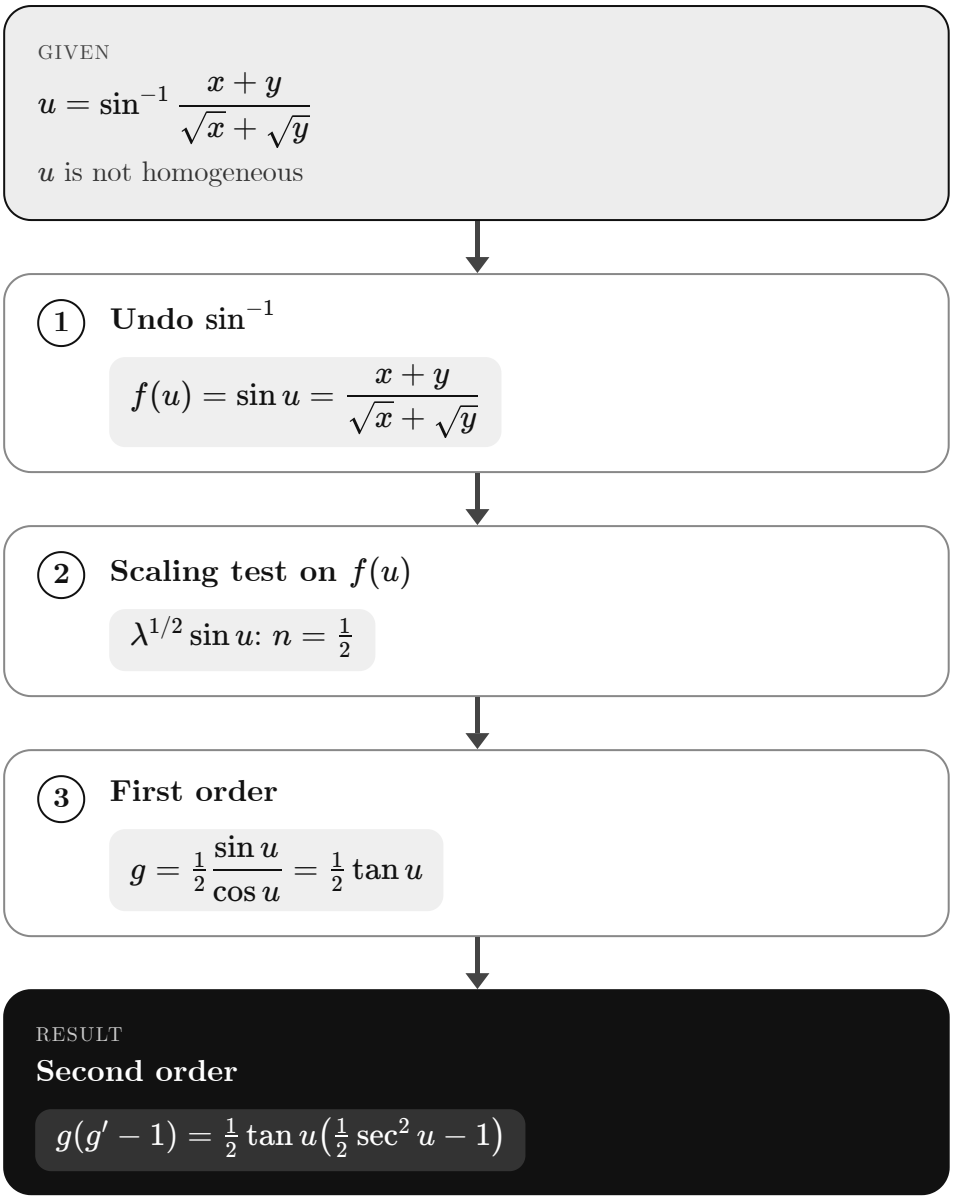


Remark. Exam questions say “using Euler's theorem”: always state the homogeneity and the degree first.

Figure F27.1. Which Euler formula? ESA ★★★

T006b 00:06:11 · T006b 00:09:08 · S006·13

28 When f(u) is homogeneous: the extension



Remark. The same four steps give $4u\log u$ for $e^{(\cdots)}$ (slide 006·12) and $1, -1$ for $\log(\dots)$ (slide 006·14).

Figure F28.1. Using the extension (slides 006·16–17) ESA ★★★

S006·16 · S006·17 · T006b 00:11:40

The extension for each outer function

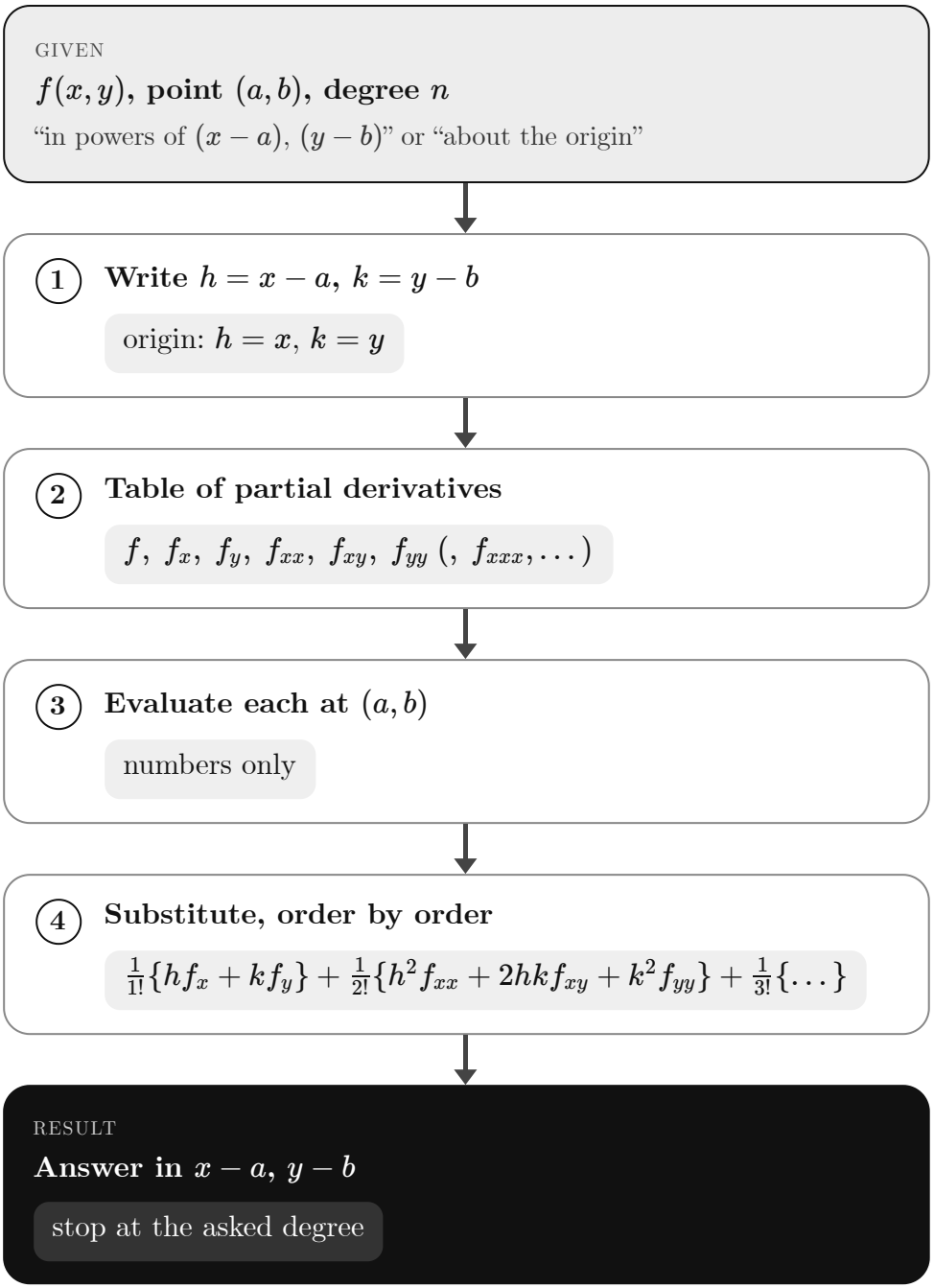
$u =$	$f(u)$	$g(u) = xu_x + yu_y$	$g(g' - 1)$	**In class 006**
$\sin^{-1}(P)$	$\sin u$	$n \tan u$	$n \tan u \, (n \sec^2 u - 1)$	$n = -3$ (slide 4); $n = \frac{1}{2}$ (slide 16)
$\cos^{-1}(P)$	$\cos u$	$-n \cot u$	$-n \cot u \, (n \csc^2 u - 1)$	$n = \frac{1}{2}$ (slide 3, u itself)
$\tan^{-1}(P)$	$\tan u$	$\frac{n}{2} \sin 2u$	$\frac{n}{2} \sin 2u \, (n \cos 2u - 1)$	$n = 2$ (ESA May/June 2024)
$\log(P)$	e^u	n	$-n$	$n = 1$ (slide 14)
e^P	$\log u$	$nu \log u$	$nu \log u \, (n \log u + n - 1)$	$n = 4$ (slide 12)

ADDED FOR CLARITY S006·4 · S006·13 · S006·15 · S006·17 · ESA May-24 1a

M6

Class 007 · Taylor's and Maclaurin's Expansion

31 Taylor's expansion of a function of two variables

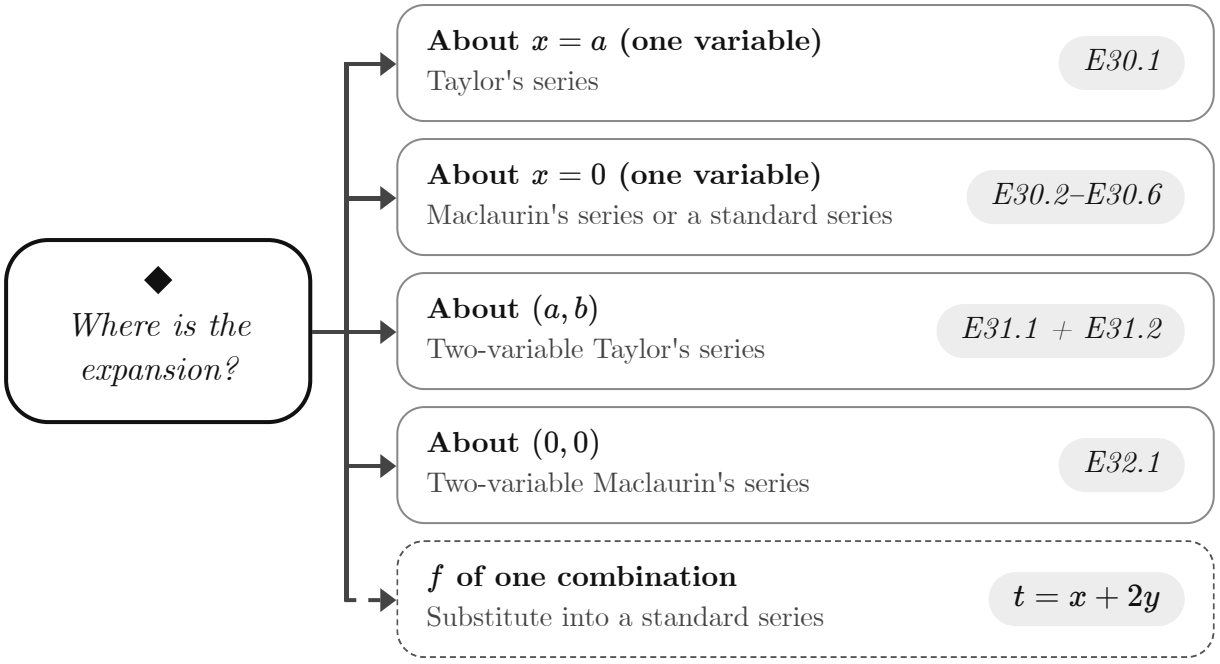


Remark. Check: if f depends only on $x + 2y$, $x - y$, ..., the one-variable series gives the same answer.

Figure F31.1. Expanding f(x, y) in Taylor's series ESA ★★★

S007·9 · ESA Jul-23 1b · ESA Dec-23 1b

32 Maclaurin's expansion of a function of two variables



Remark. The dashed route is a shortcut added for clarity; it is also the quickest way to check an answer.

Figure F32.1. Which series? ESA ★★

T007 00:11:17 · S007·9 · S007·10

Taylor's and Maclaurin's series side by side

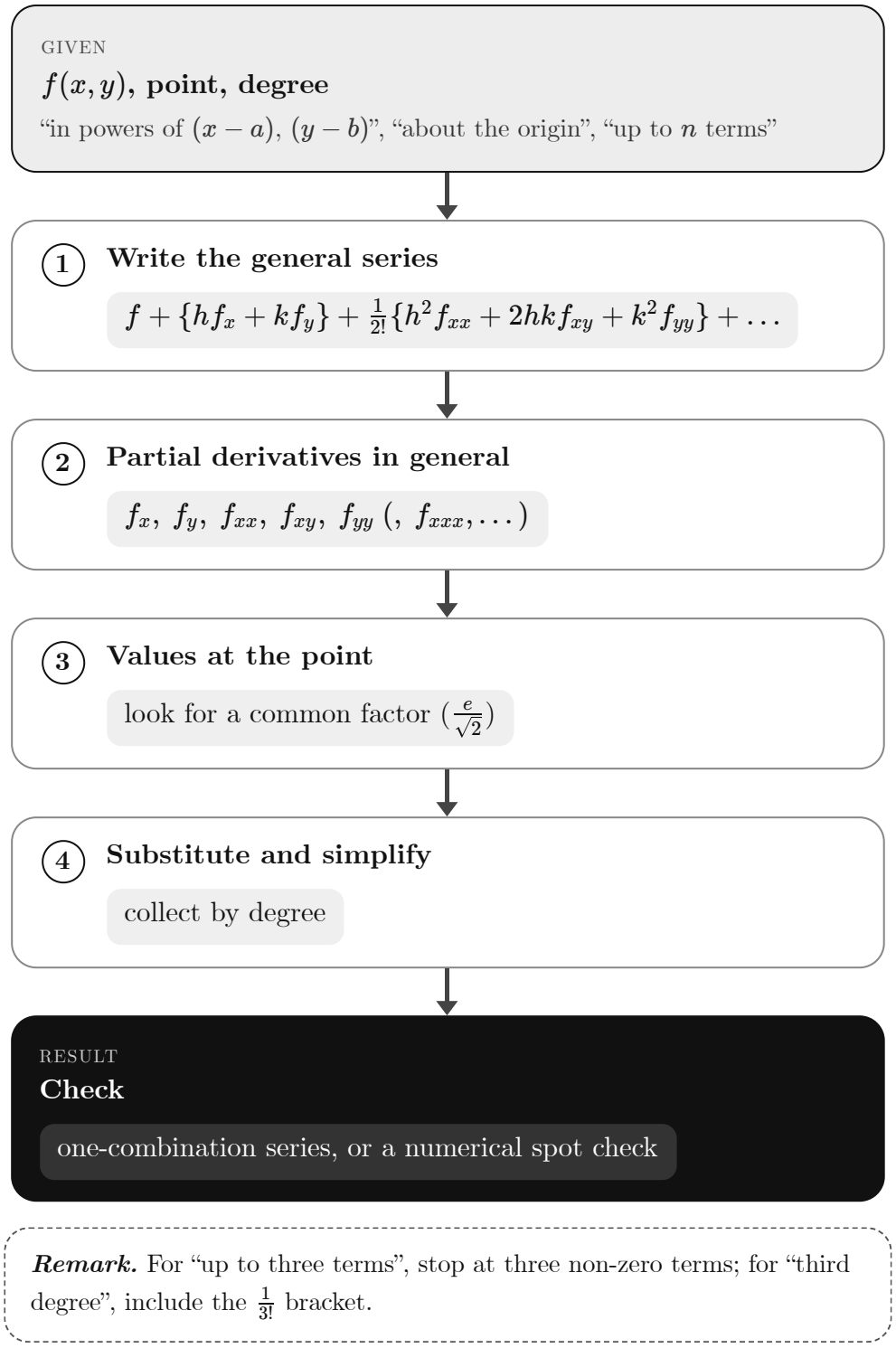
	One variable	Two variables
Taylor (about a point)	$f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$	$f(a,b) + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots$
Maclaurin (about the origin)	$f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$	$f(0,0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \dots$
First degree	Tangent line	Tangent plane
Coefficients at order n	$\frac{1}{n!}$	$\frac{1}{n!}\times$ binomial (1,2,1; 1,3,3,1)
Condition	f and derivatives finite at the point	f and all partials finite at the point

ADDED FOR CLARITY S007·4 · S007·9 · S007·10

M7

Class 008 · Problems on Taylor's and Maclaurin's Series

33 Taylor expansion about a point (a, b)



Remark. For "up to three terms", stop at three non-zero terms; for "third degree", include the $\frac{1}{3!}$ bracket.

Figure F33.1. Solving a Taylor or Maclaurin expansion problem ESA ★★★

T008 00:00:21 · T008 00:02:07 · T008 00:05:19

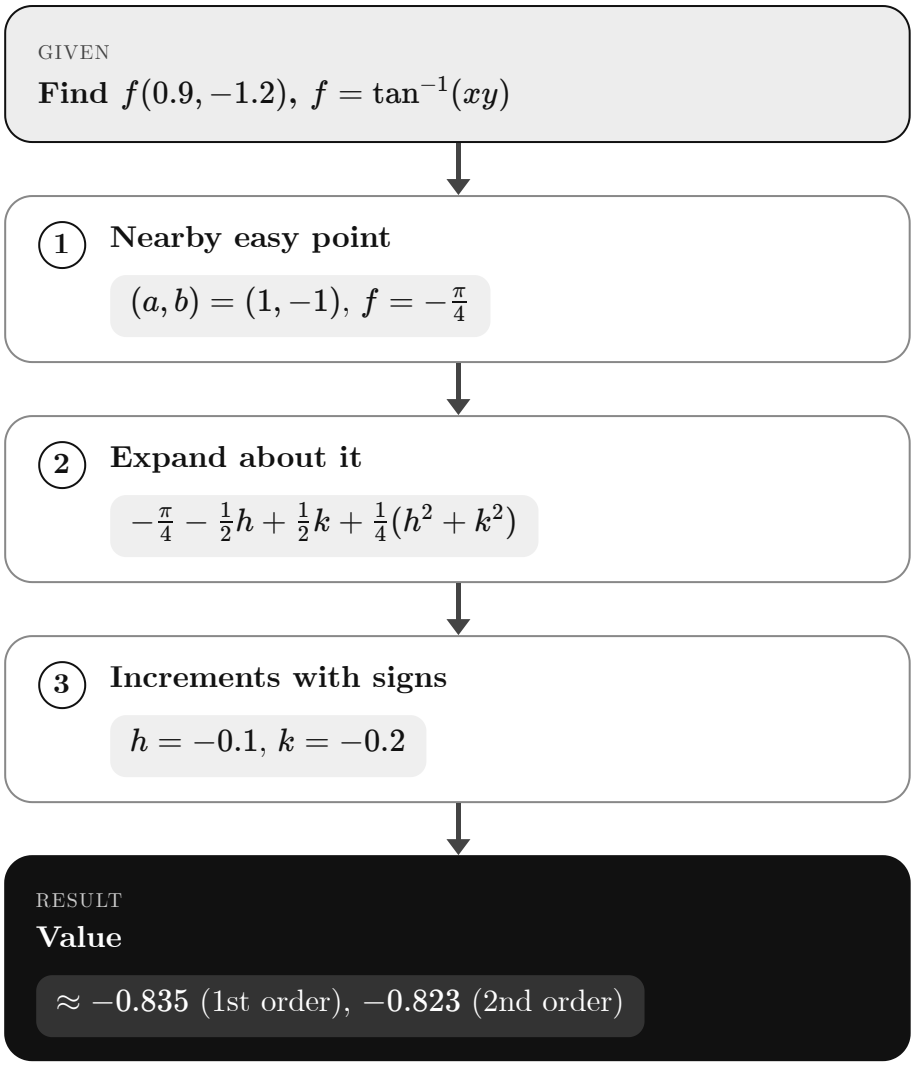
34 Maclaurin expansion about the origin

Quick checks with a one-variable series

Function	Combination t	One-variable series	Problem
$\log(2x+y+1)$	$2x+y$	$t-\frac{t^2}{2}+\frac{t^3}{3}-\dots$	W34.1
e^{x-y}	$x-y$	$1+t+\frac{t^2}{2}+\frac{t^3}{6}+\dots$	W34.2
$\log(1+x-y)$	$x-y$	$t-\frac{t^2}{2}+\frac{t^3}{3}-\dots$	W34.4
$\sin(x+2y)$	$x+2y$	$t-\frac{t^3}{6}+\dots$	W34.6
$e^x\log(1+y)$, $e^{ax}\sin by$	product	multiply the two series, keep degree ≤ 3	W34.3, W34.5

added for clarity S008·6 · S008·11 · ESA Jan-24 s1 1b · ESA Jun-24 1c

35 Approximate values from a Taylor polynomial



Remark. Exact value -0.8238: the second-order term is worth keeping.

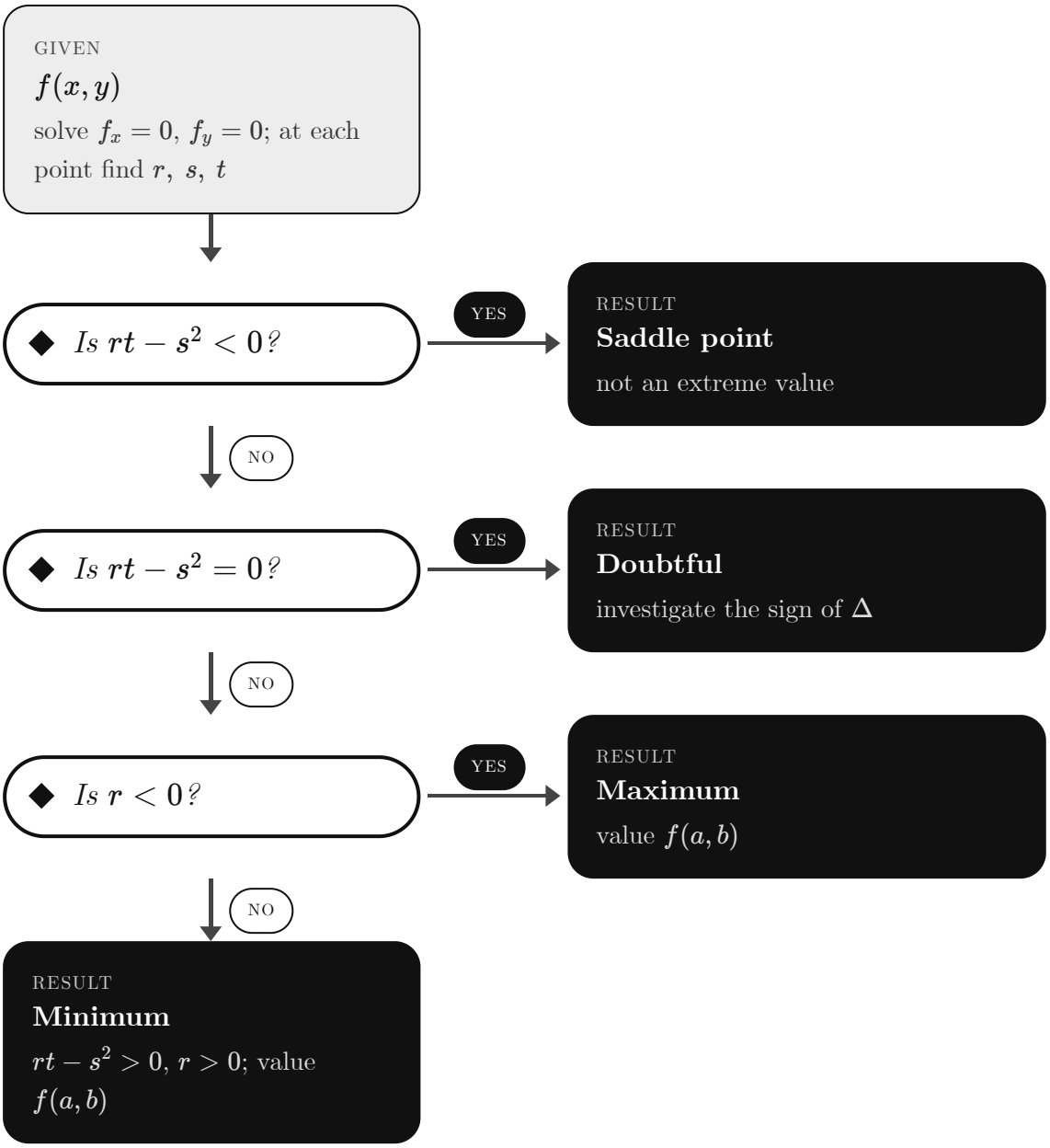
Figure F35.1. Approximate value from a Taylor polynomial ESA ★★

S008·7 · S008·9 · T008 00:12:10

M8

Class 009 · Maxima and Minima of a Function of Two Variables

38 Conditions and the working rule



Remark. Repeat the ladder for every stationary point.

Exam trap. Never classify from r alone.

Figure F38.1. Working rule: nature of a stationary point ESA ★★★

S009·8 · S009·9 · T009 00:11:32

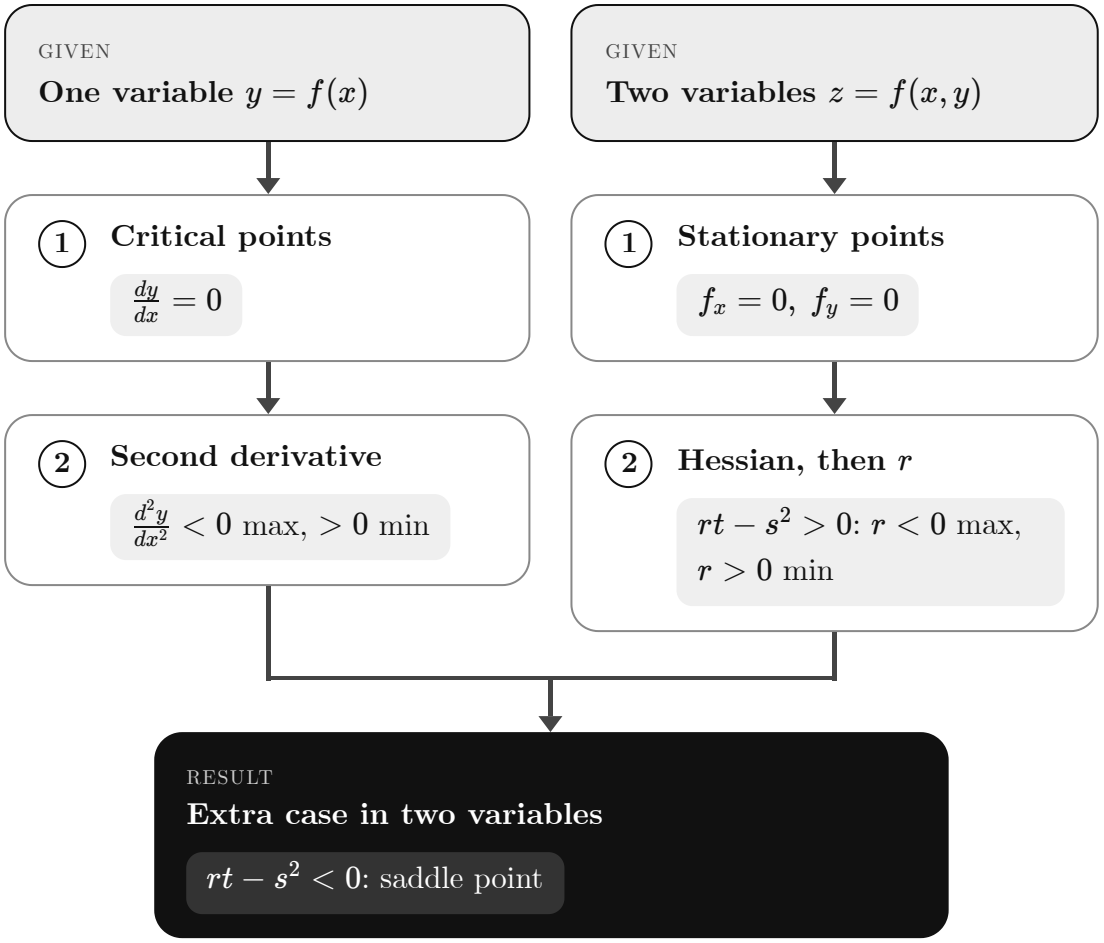


Figure F38.2. From one variable to two ESA ★★

T009 00:06:00 · T009 00:10:20

The four outcomes of the test

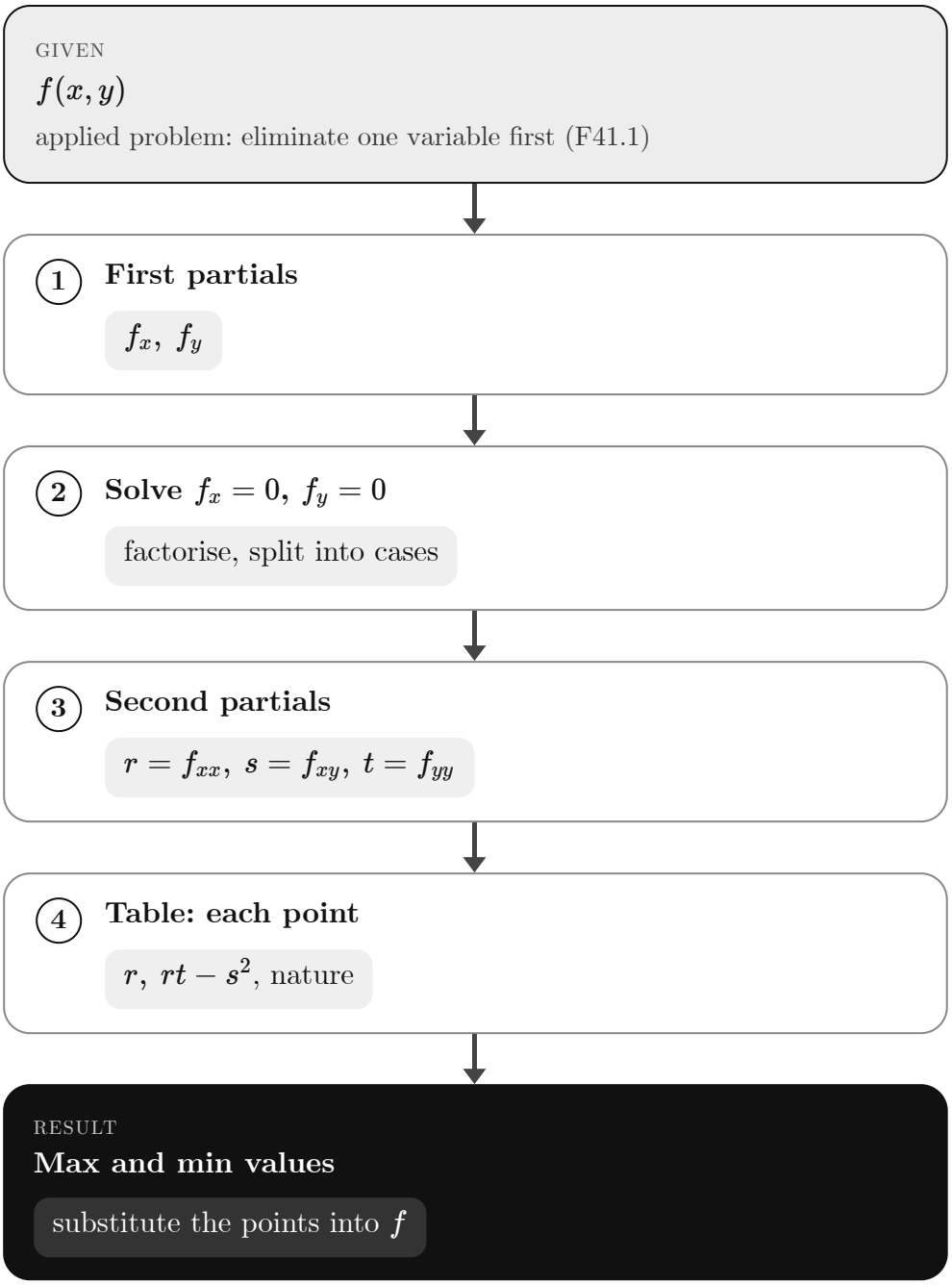
$rt - s^2$	r	Nature of (a, b)	Example at the origin
> 0	< 0	Maximum	$-x^2 - y^2$
> 0	> 0	Minimum	$x^2 + y^2$ (W38.1)
< 0	any	Saddle point	$x^2 - y^2$ (W38.2)
$= 0$	any	Doubtful: further investigation	$x^4 + y^4$ (min), $x^4 - y^4$ (neither)

S009·7 · S009·8 · S009·9

M9

Class 010 · Problems on Maxima and Minima

39 The method, and the ESA cubic functions



Remark. Write a table with one row per stationary point: point, r, s, t, $rt - s^2$, nature, value.

Figure F39.1. Solving a maxima–minima problem ESA ★★★

S010·7 · S010·8 · T010b 00:05:00

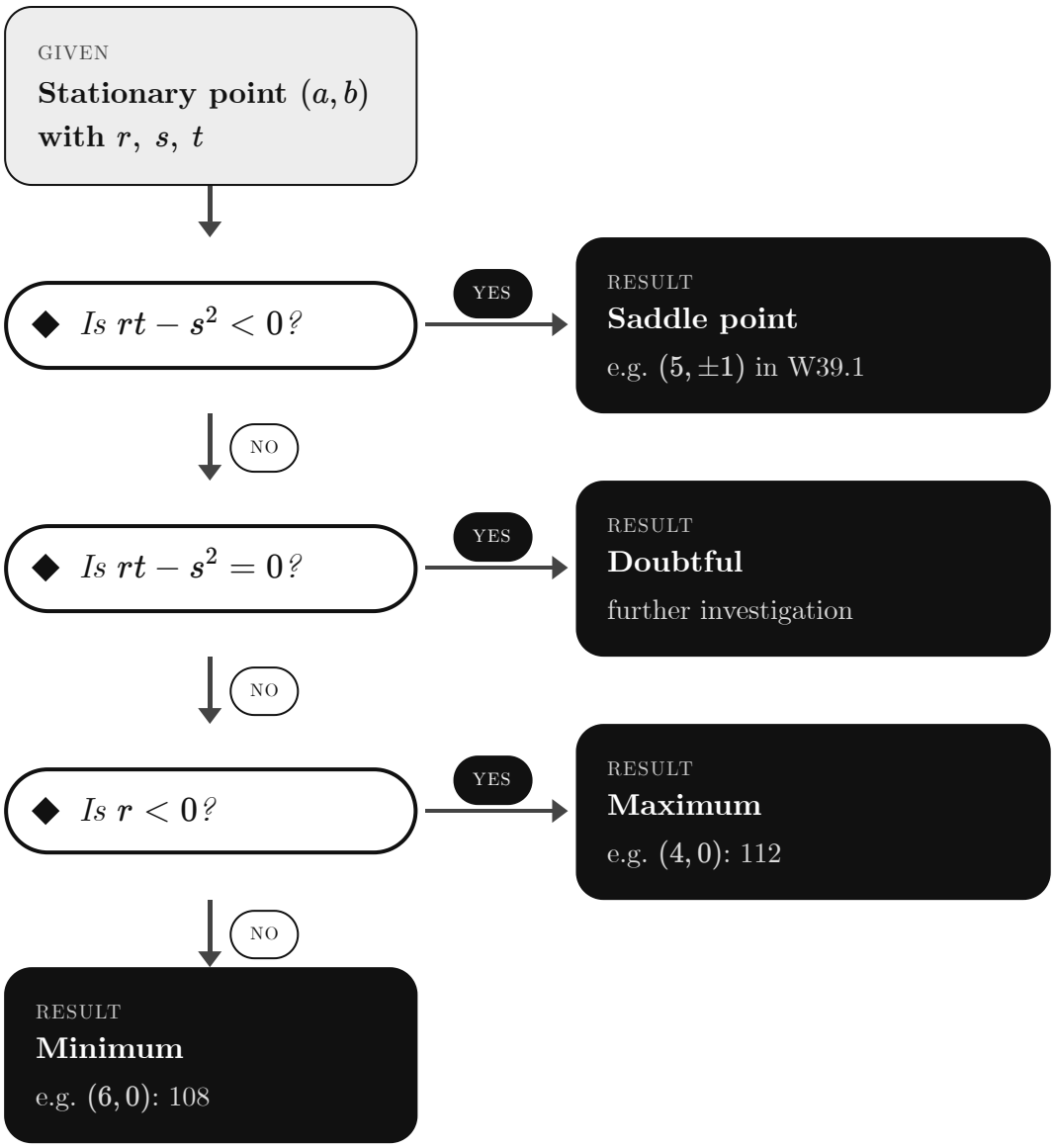


Figure F39.2. Classifying each stationary point ESA ★★★

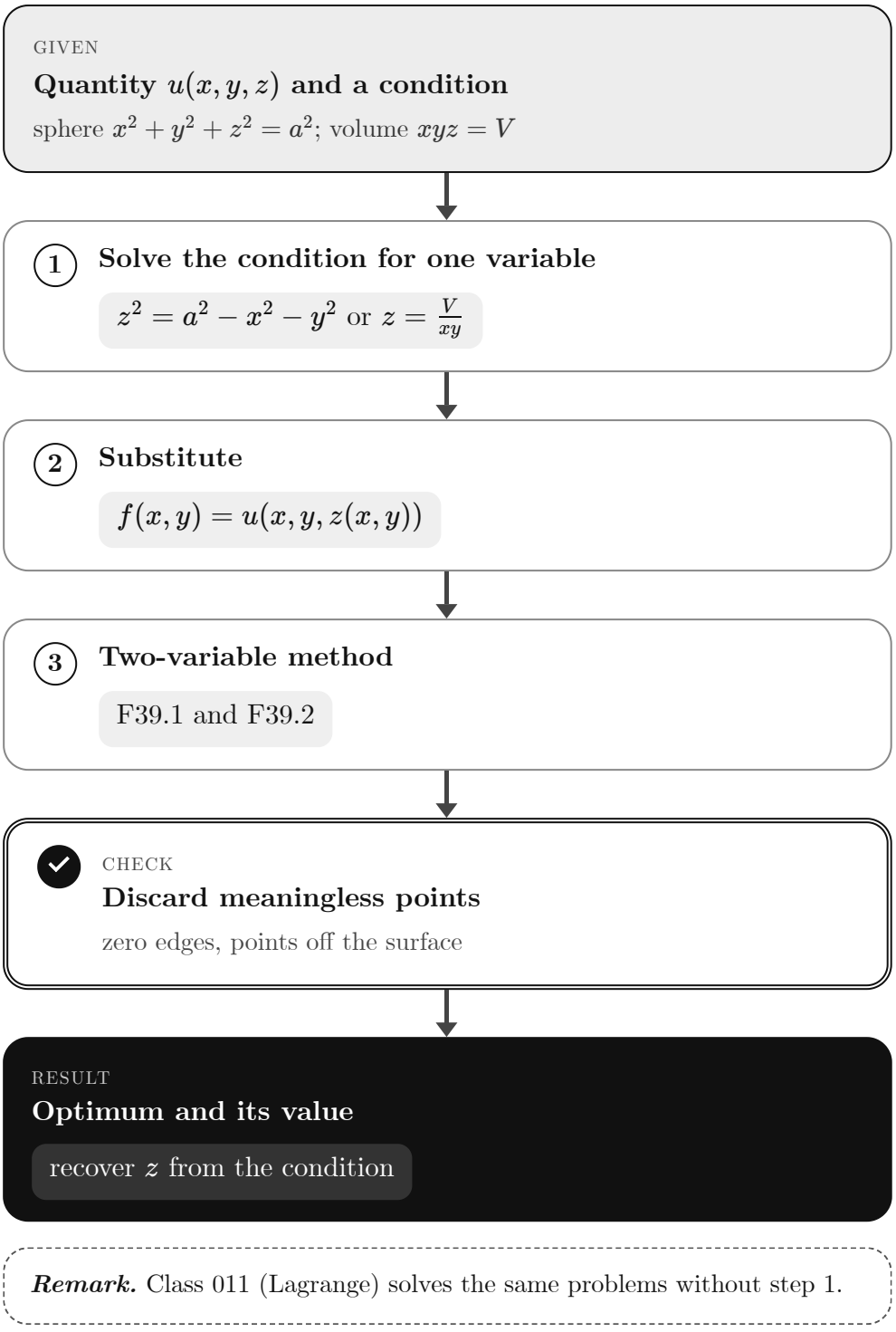
S010·8 · T010b 00:04:02

Stationary points of the two ESA functions

Function	Point	r	$rt - s^2$	Nature	f
$x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$	$(4, 0)$	-6	36	Maximum	112
	$(6, 0)$	6	36	Minimum	108
	$(5, \pm 1)$	0	-36	Saddle	—
$x^3 - 12x + y^3 + 3y^2 - 9y$	$(-2, -3)$	-12	144	Maximum	43
	$(2, 1)$	12	144	Minimum	-21
	$(2, -3)$, $(-2, 1)$	± 12	-144	Saddle	—

S010·8 · ESA Dec-23 1c · ESA Jan-24 s2 1a

41 Applied problems: eliminate one variable first



Remark. Class 011 (Lagrange) solves the same problems without step 1.

Figure F41.1. Applied problem by elimination ESA ★★

S010·3 · S010·5 · T010a 00:00:40 · T010a 00:08:27

M10

Class 011 · Lagrange's Method of Undetermined Multipliers

42 Lagrange's method: the auxiliary function and the procedure

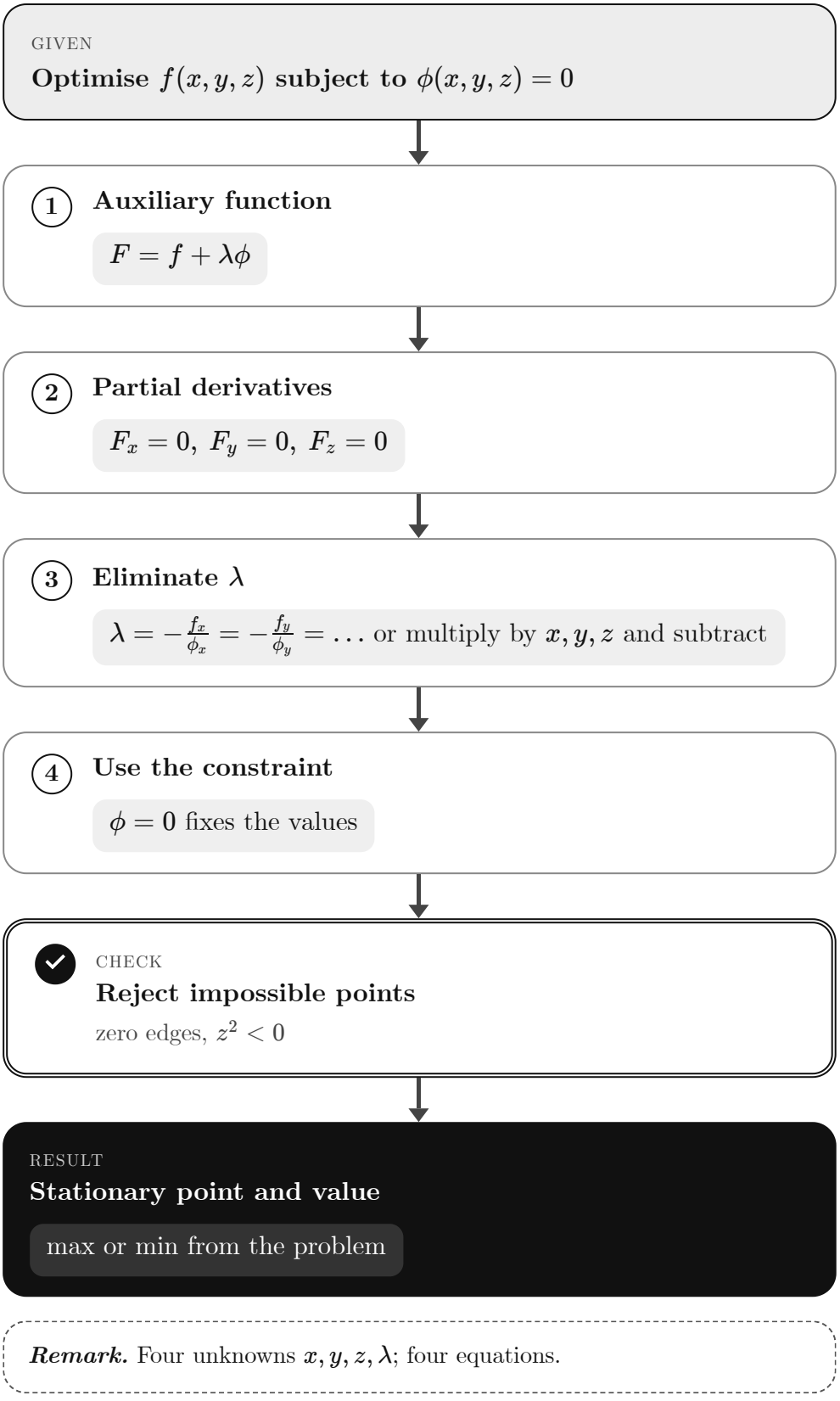


Figure F42.1. Lagrange's working procedure ESA ★★★

S011·3 · T011a 00:01:18

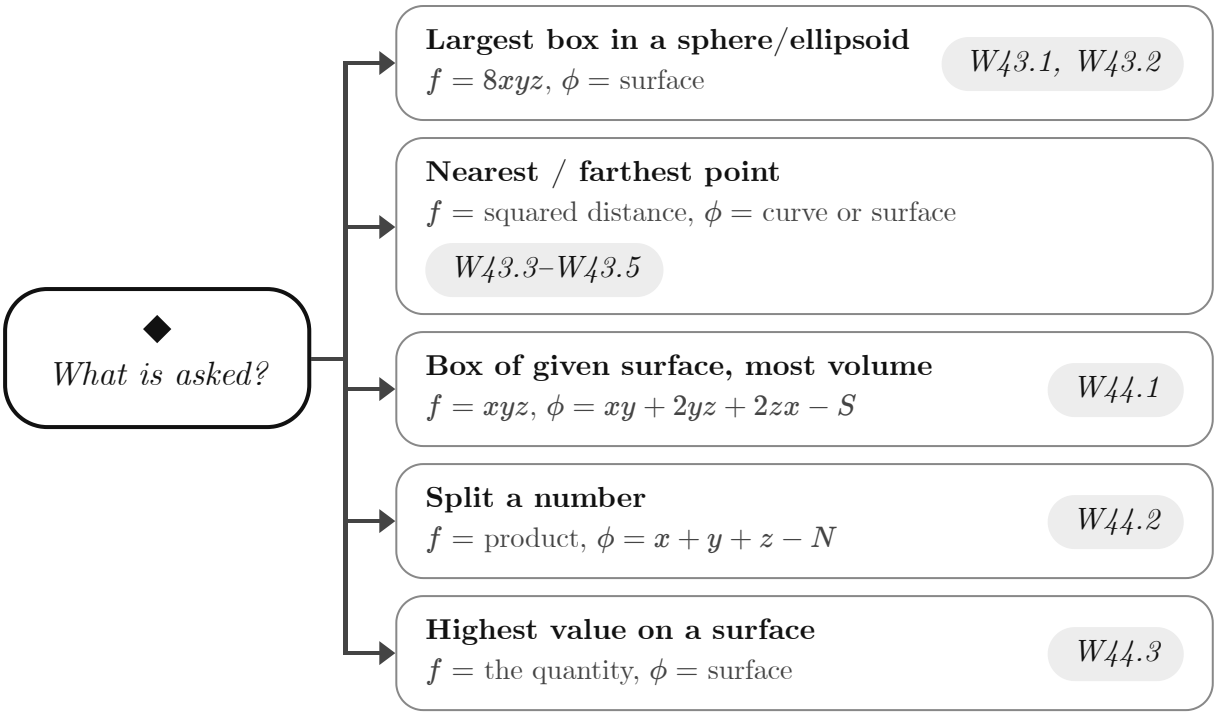


Figure F42.2. Choosing f and ϕ ESA ★★★

T011a 00:09:15 · T011b 00:01:46 · S011·10 · S011·12

Elimination (class 010) and Lagrange's method

	Elimination	Lagrange's multipliers
Constraint used	Solved for one variable and substituted	Added as $\lambda\phi$
Equations	$f_x = f_y = 0$ in two variables	$F_x = F_y = F_z = 0$, $\phi = 0$
Nature of the point	$rt - s^2$ and r decide	Not decided; from the problem
More variables / constraints	Hard	Extends directly
Temperature $400xyz^2$	50 (class 010)	50 (W44.3)

S011·4 · T011a 00:01:36 · T011a 00:04:23